

BFS Traversal

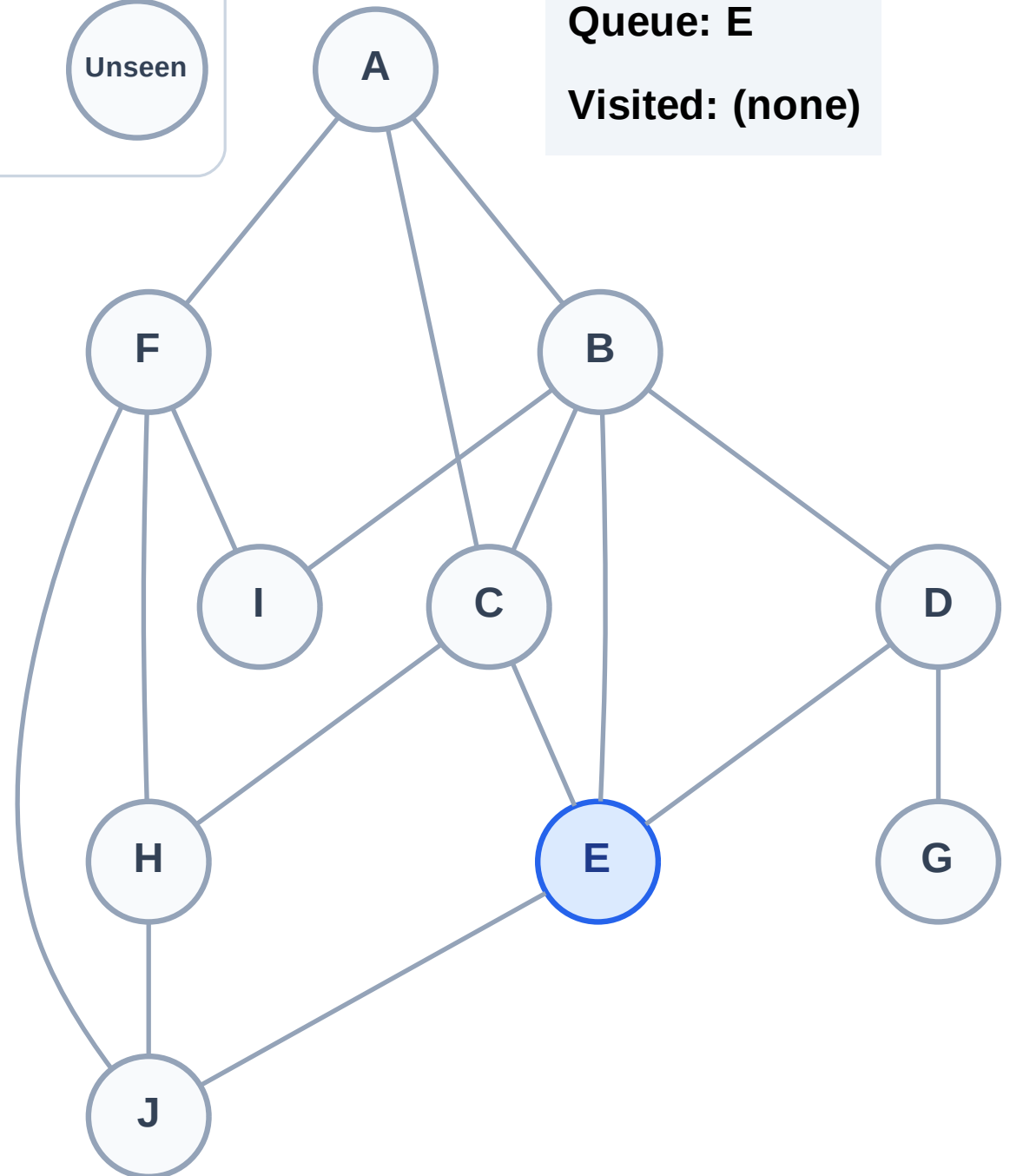
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

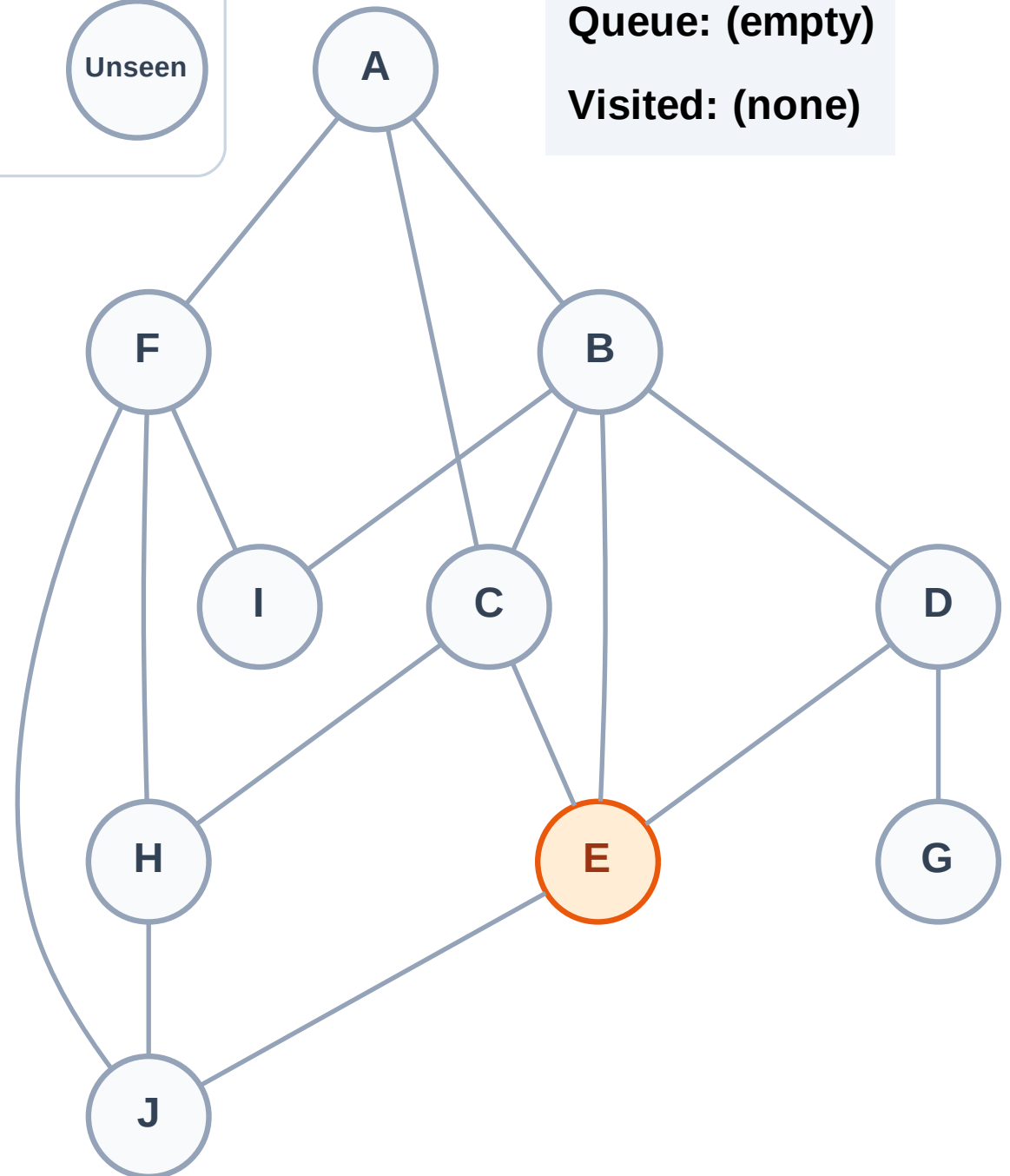
Step 2: Process node E

Legend



Queue: (empty)

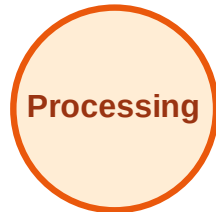
Visited: (none)



BFS Traversal

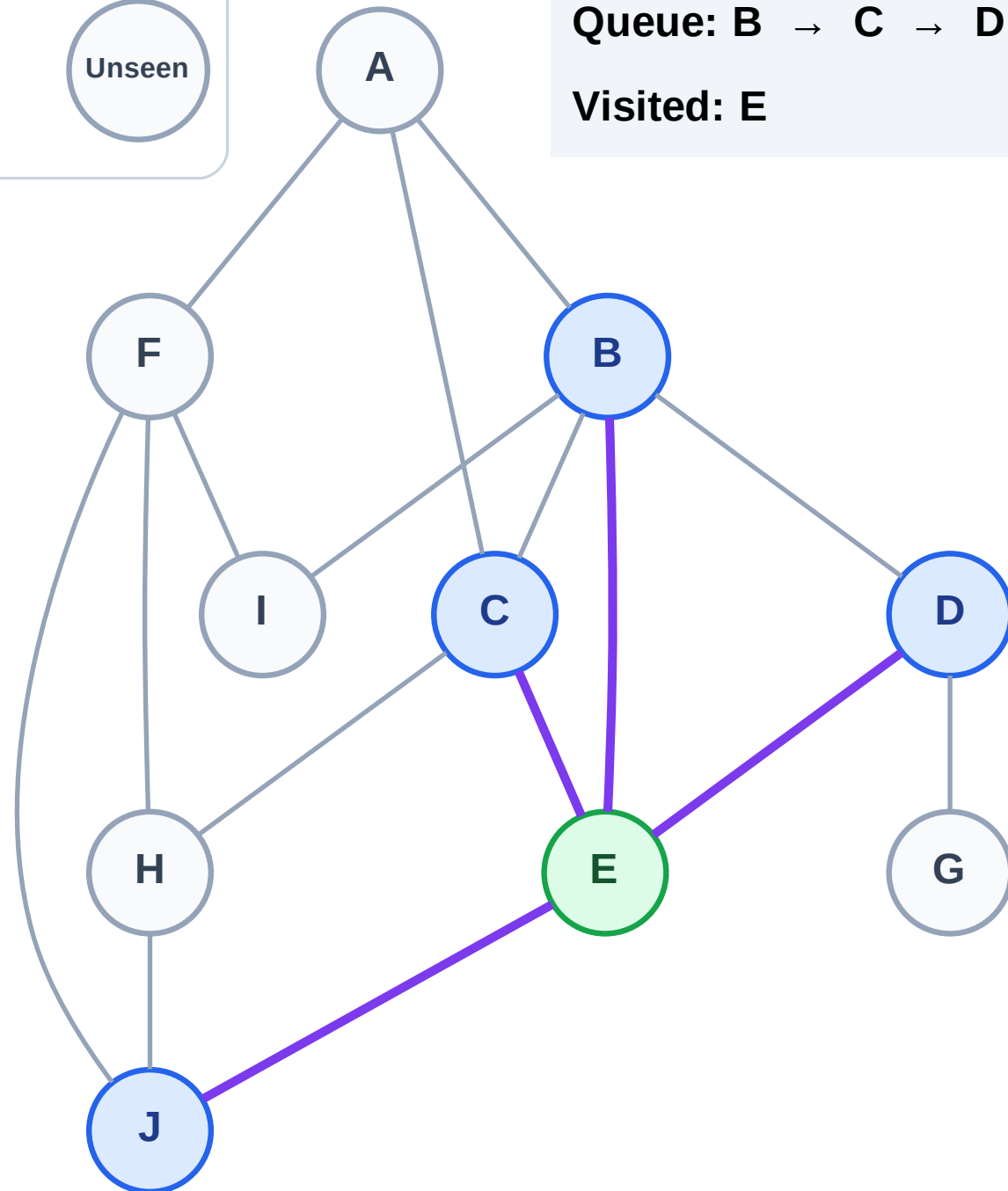
Step 3: Discover B, C, D, J from E

Legend



Queue: B → C → D → J

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

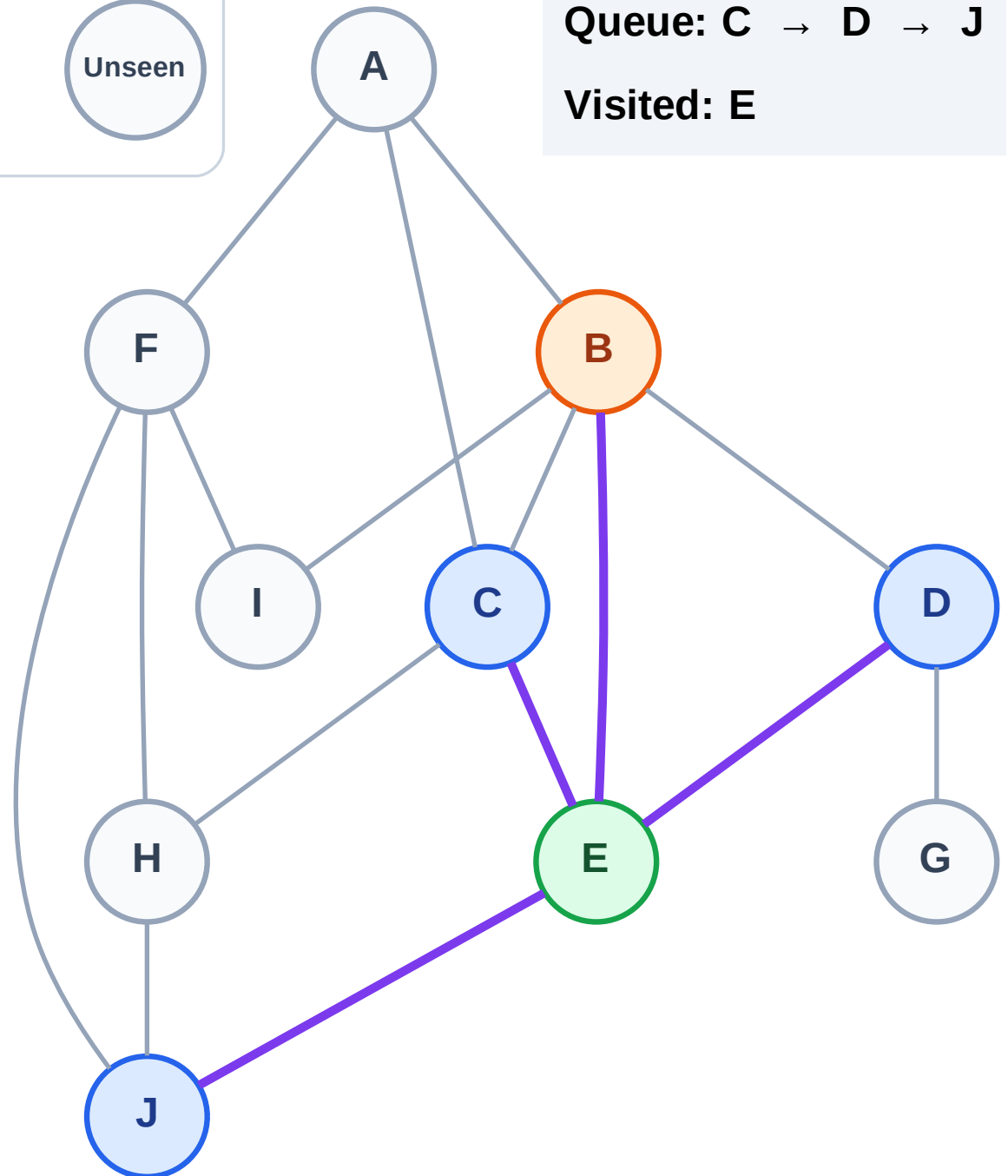
Processing

Discovered

Unseen

Queue: C → D → J

Visited: E



BFS Traversal

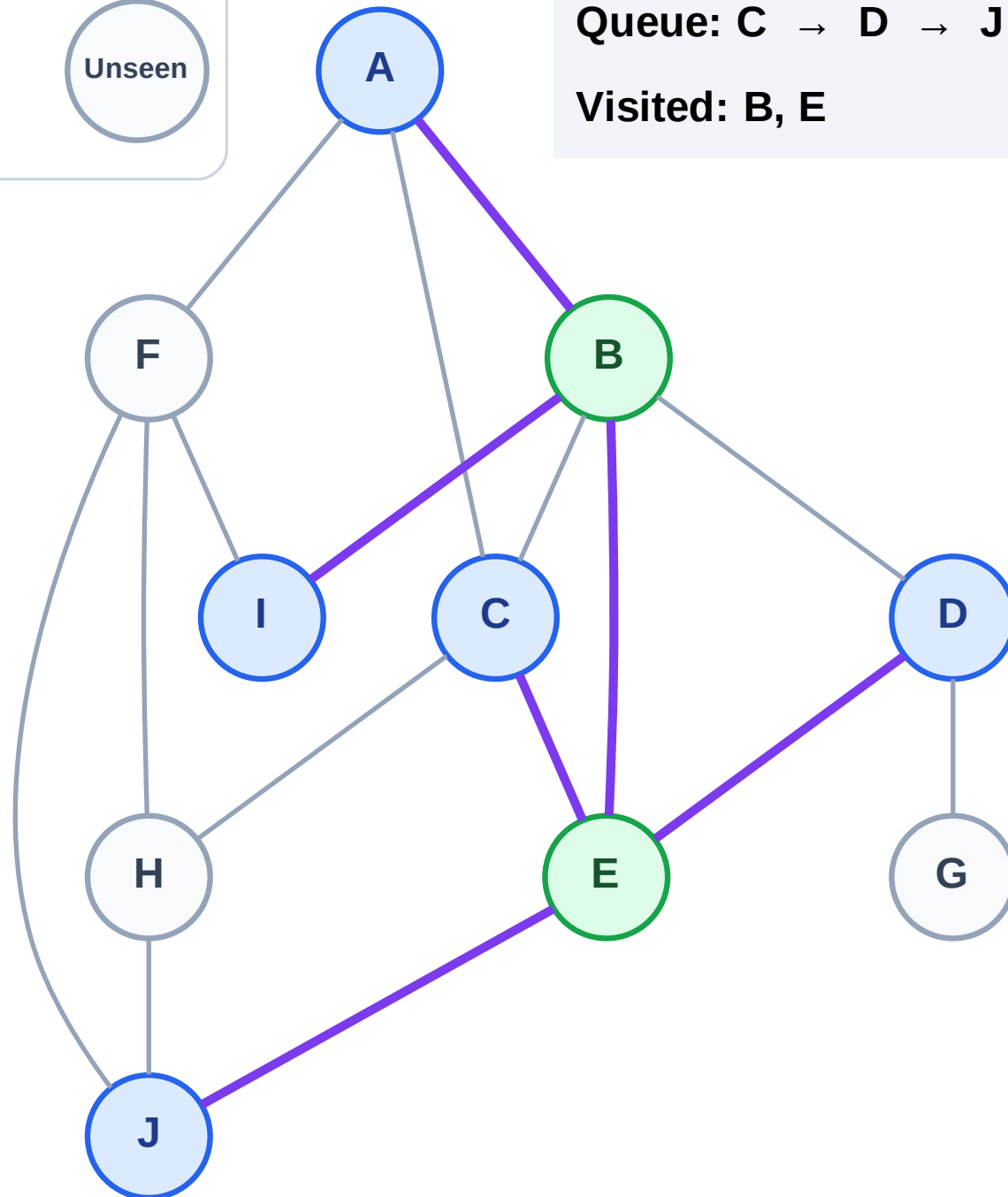
Step 5: Discover A, I from B

Legend



Queue: C → D → J → A → I

Visited: B, E



BFS Traversal

Step 6: Process node C

Legend

Complete

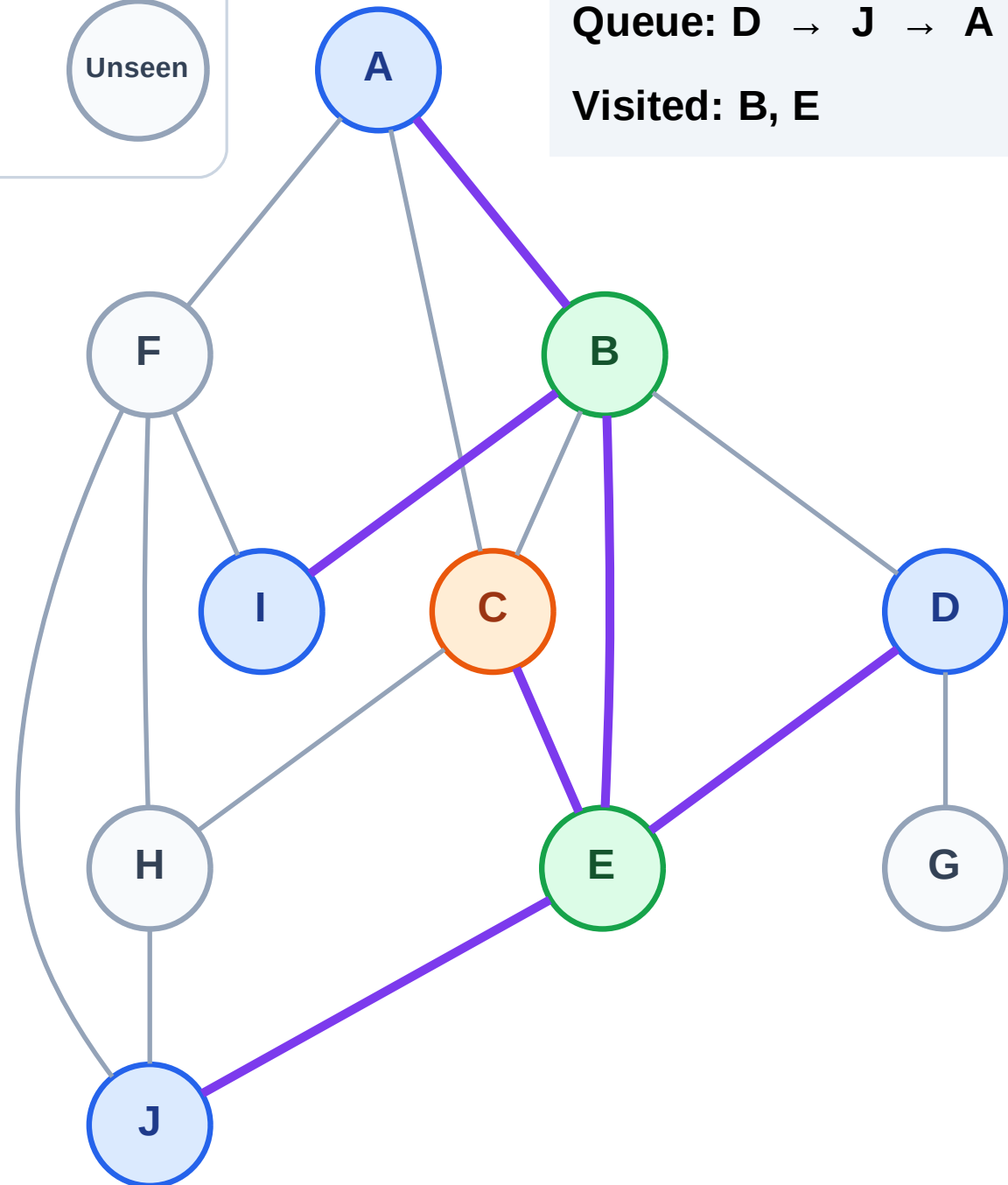
Processing

Discovered

Unseen

Queue: D → J → A → I

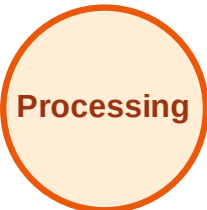
Visited: B, E



BFS Traversal

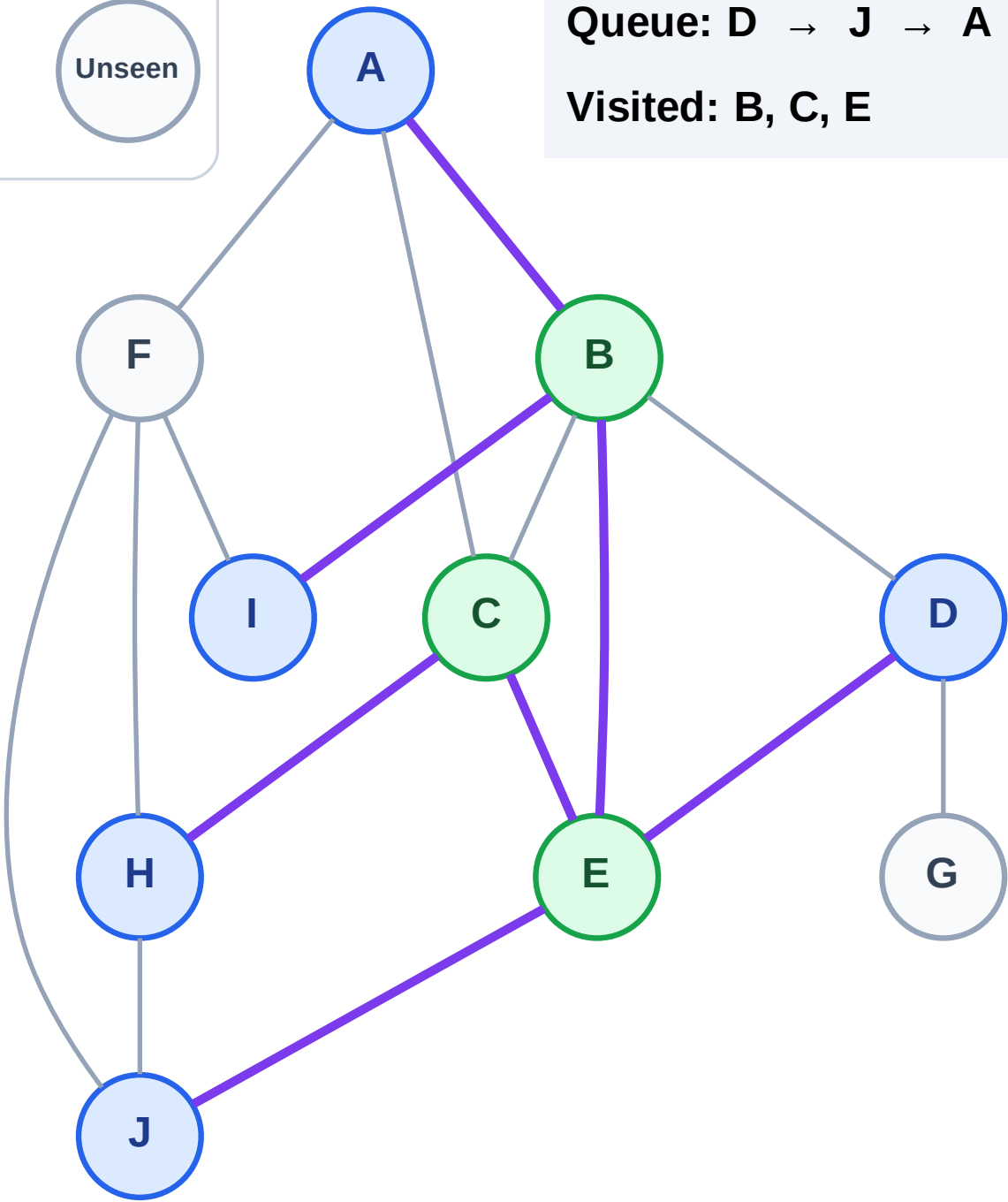
Step 7: Discover H from C

Legend



Queue: D → J → A → I → H

Visited: B, C, E



BFS Traversal

Step 8: Process node D

Legend

Complete

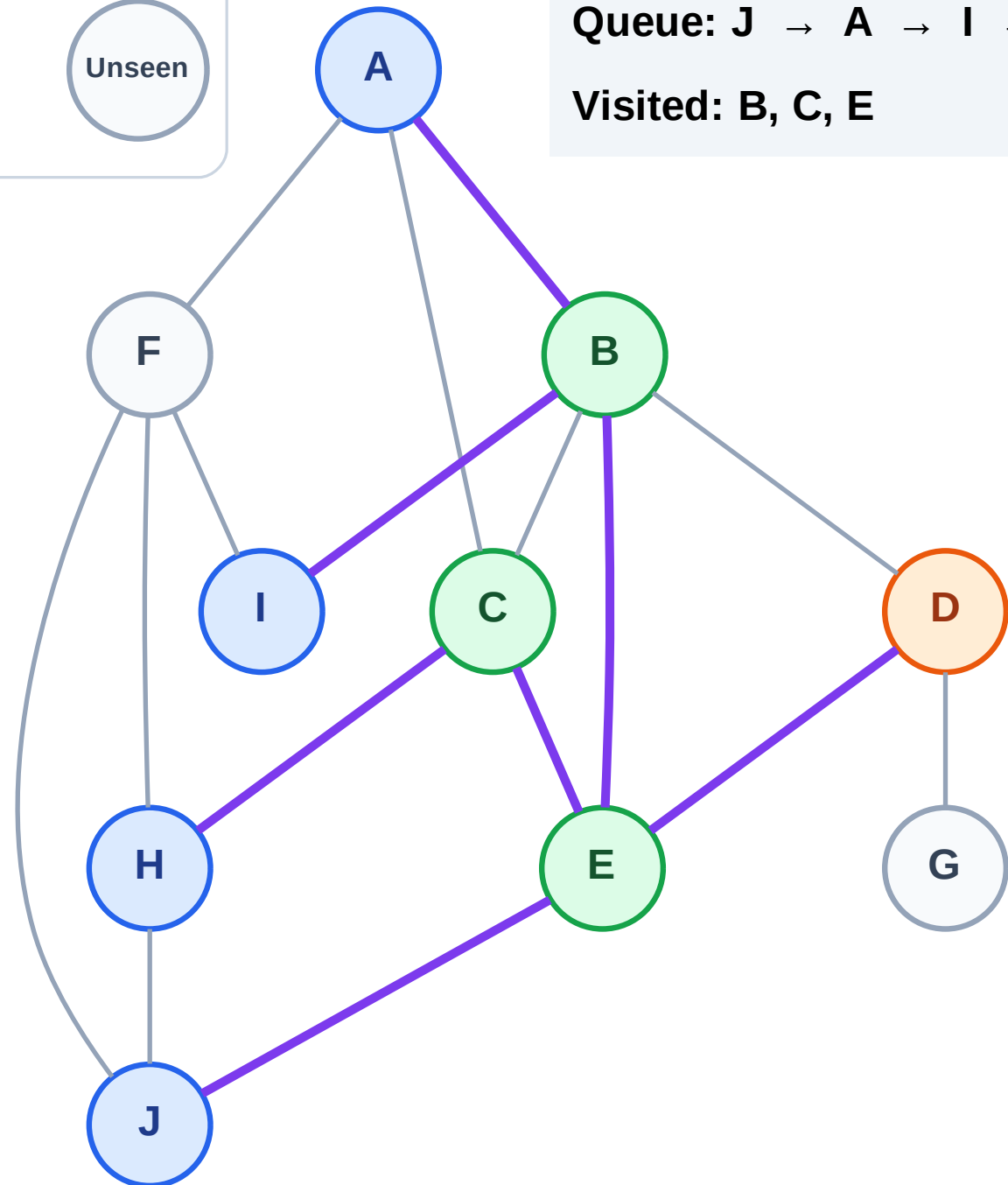
Processing

Discovered

Unseen

Queue: J → A → I → H

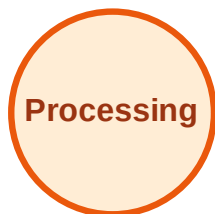
Visited: B, C, E



BFS Traversal

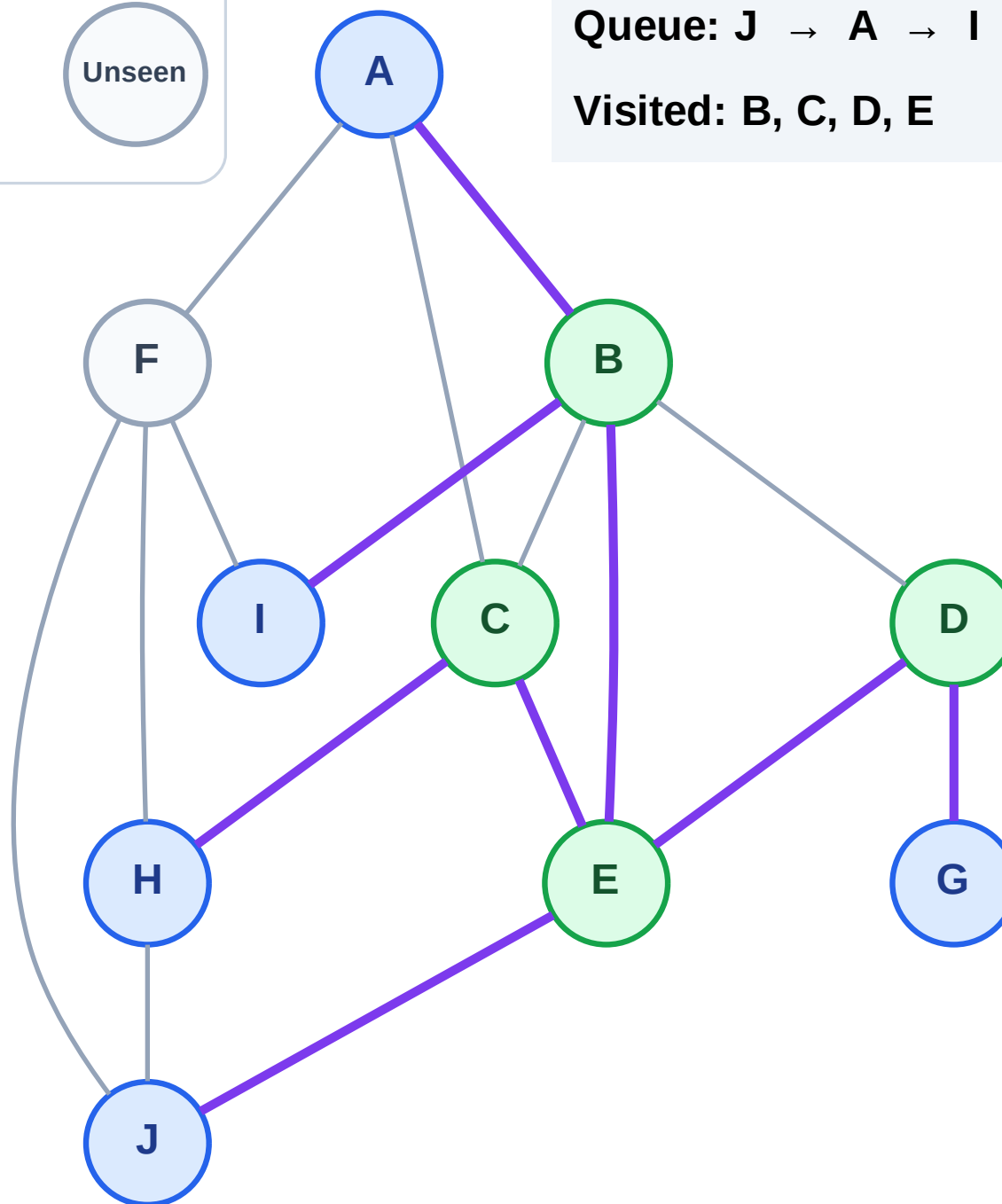
Step 9: Discover G from D

Legend



Queue: J → A → I → H → G

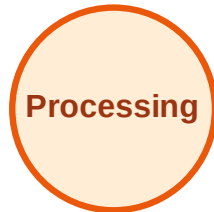
Visited: B, C, D, E



BFS Traversal

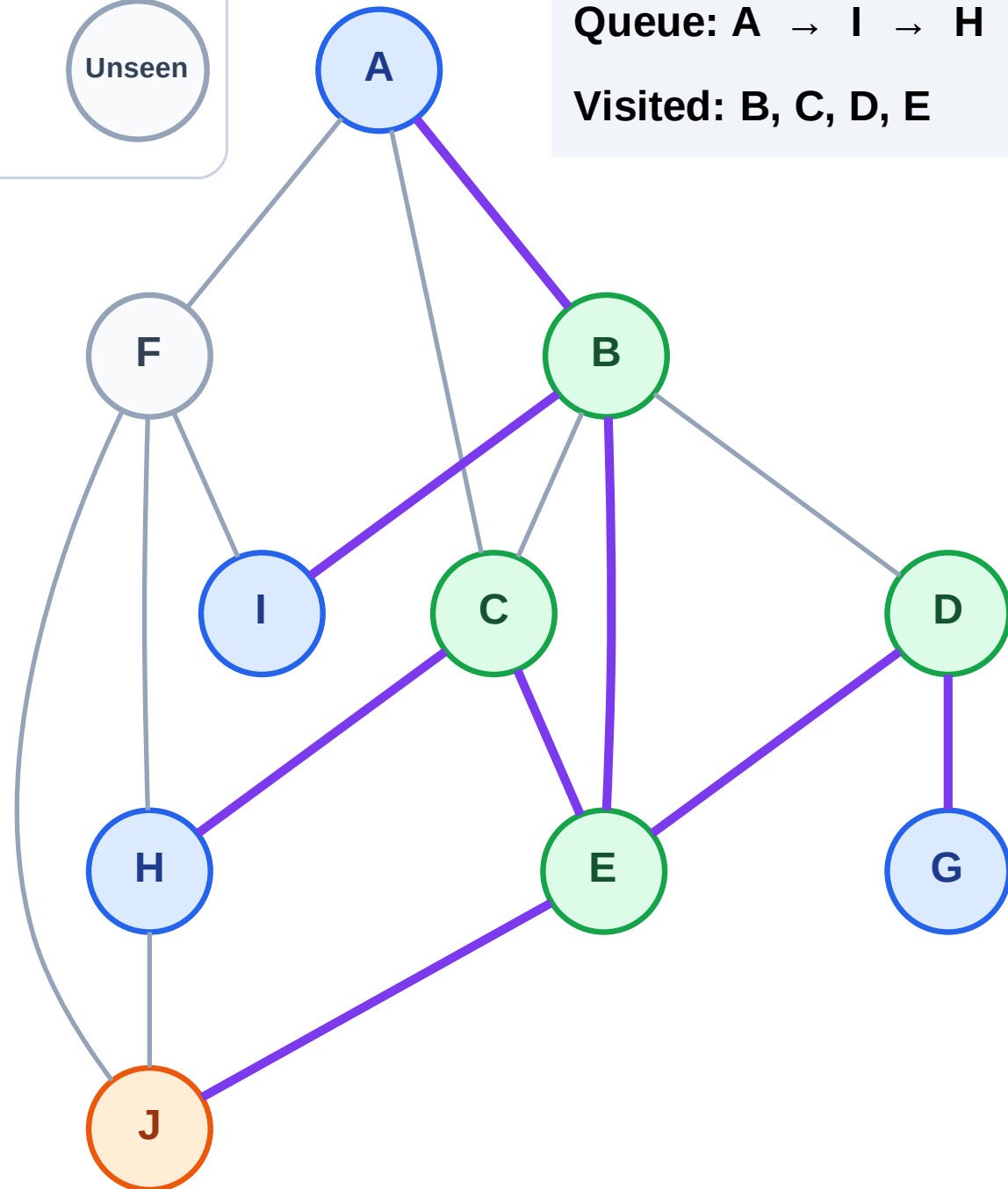
Step 10: Process node J

Legend



Queue: A → I → H → G

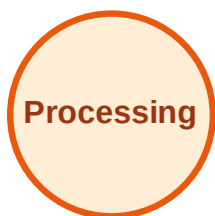
Visited: B, C, D, E



BFS Traversal

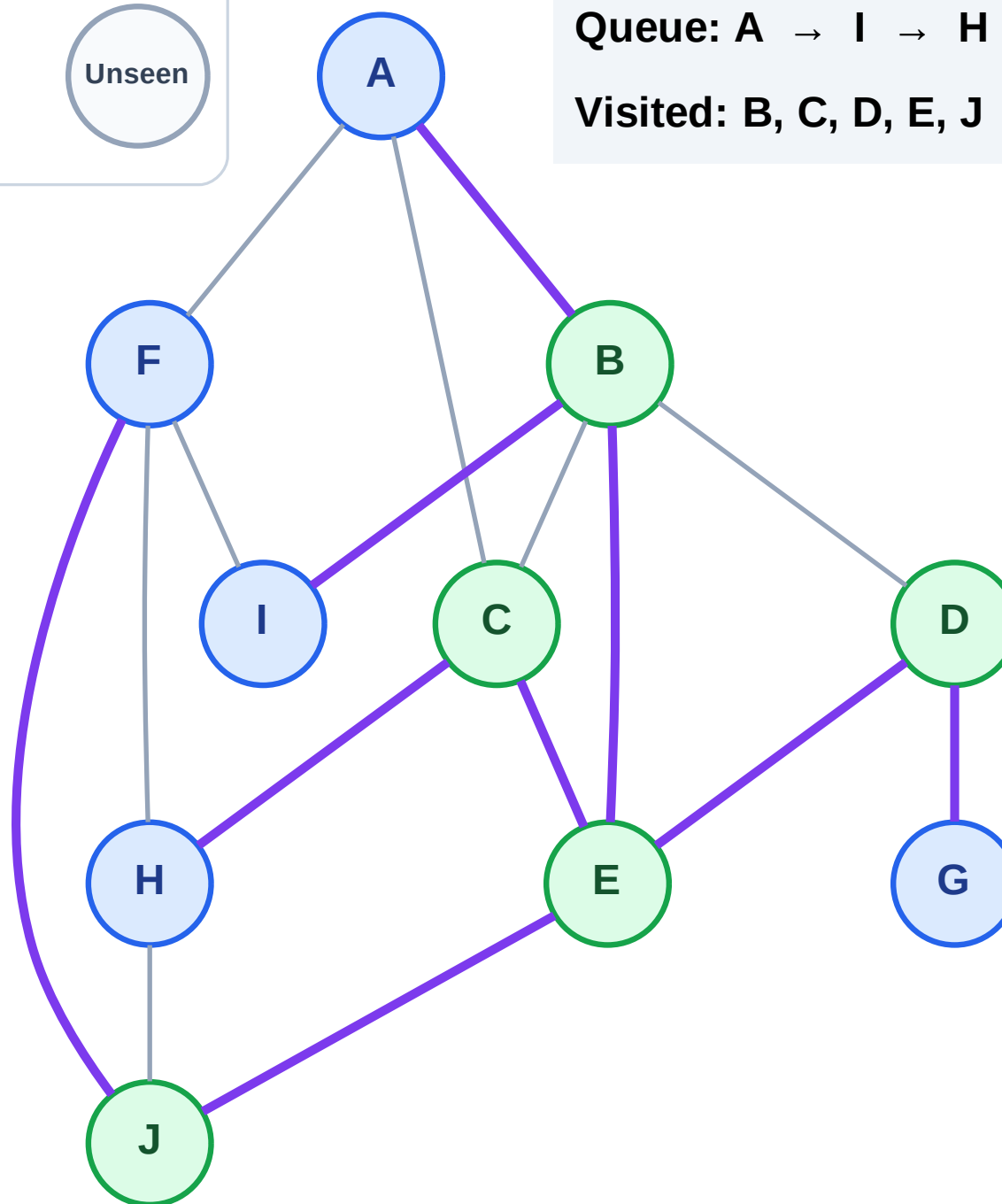
Step 11: Discover F from J

Legend



Queue: A → I → H → G → F

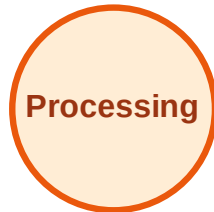
Visited: B, C, D, E, J



BFS Traversal

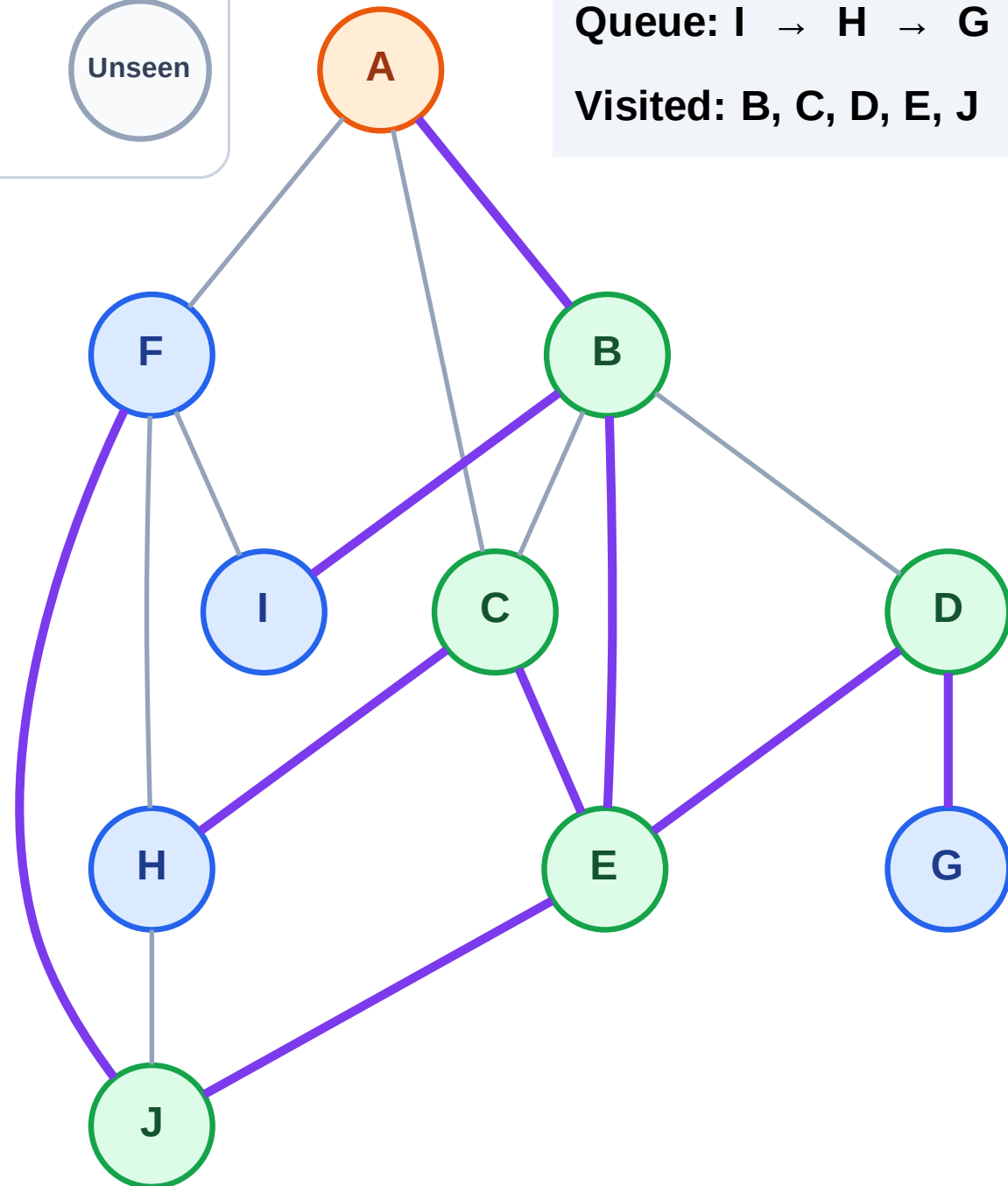
Step 12: Process node A

Legend



Queue: I → H → G → F

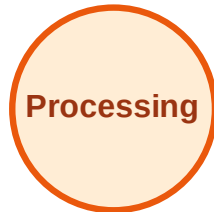
Visited: B, C, D, E, J



BFS Traversal

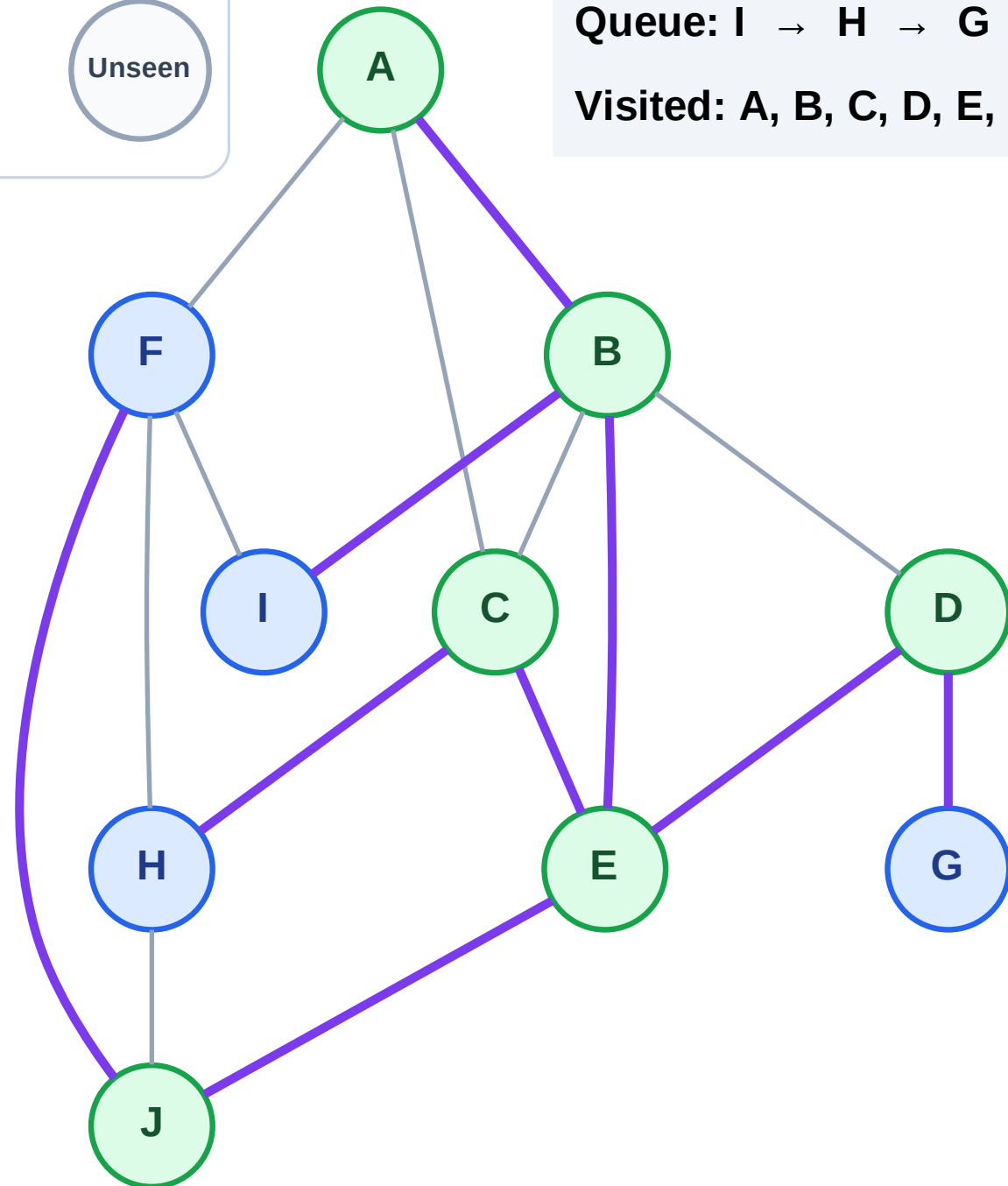
Step 13: No new neighbors from A

Legend



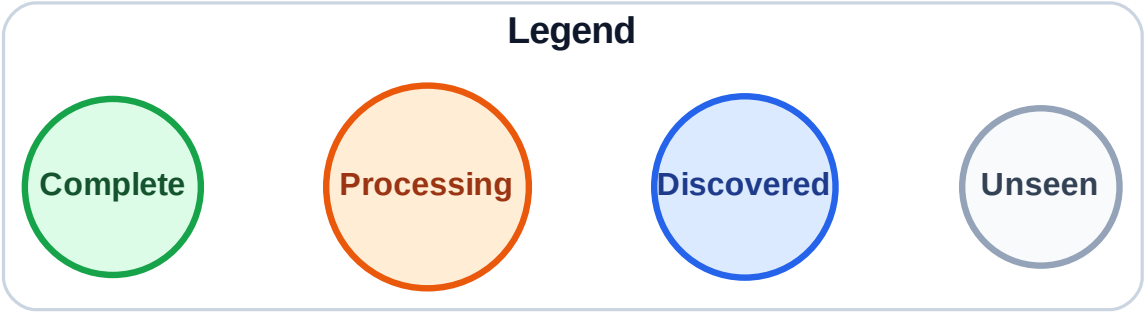
Queue: I → H → G → F

Visited: A, B, C, D, E, J

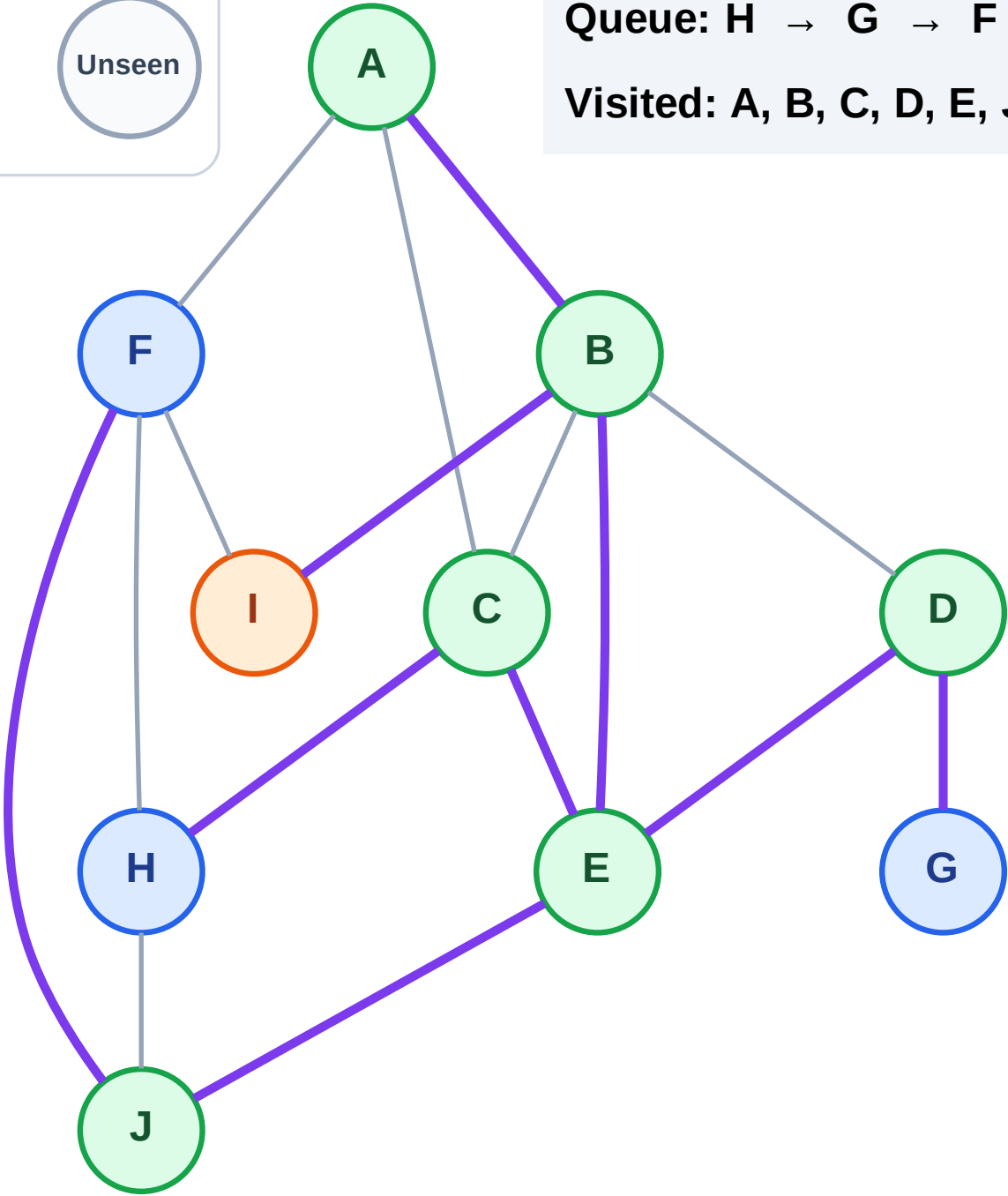


BFS Traversal

Step 14: Process node I



Queue: H → G → F
Visited: A, B, C, D, E, J



BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

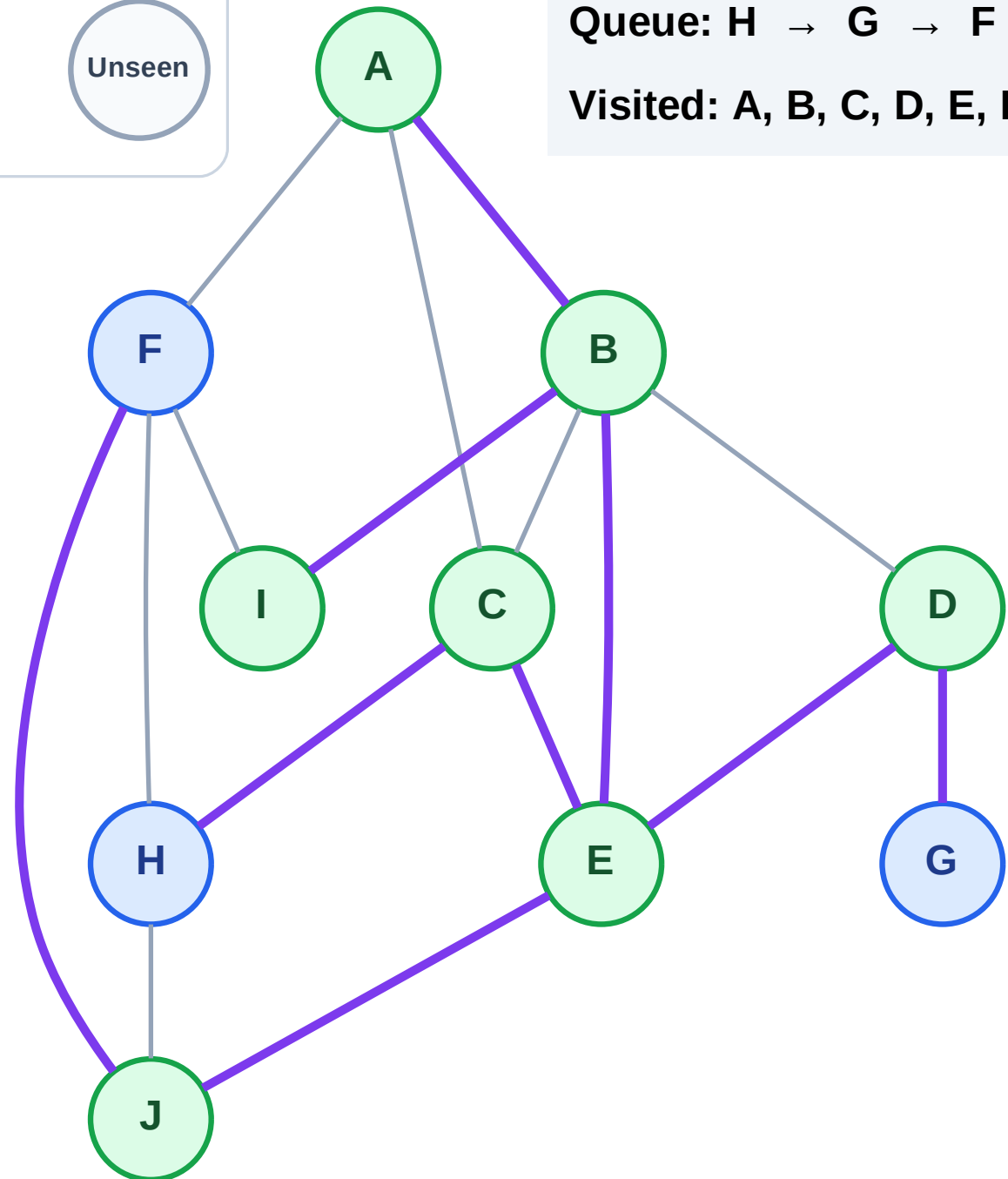
Processing

Discovered

Unseen

Queue: H → G → F

Visited: A, B, C, D, E, I, J



BFS Traversal

Step 16: Process node H

Legend

Complete

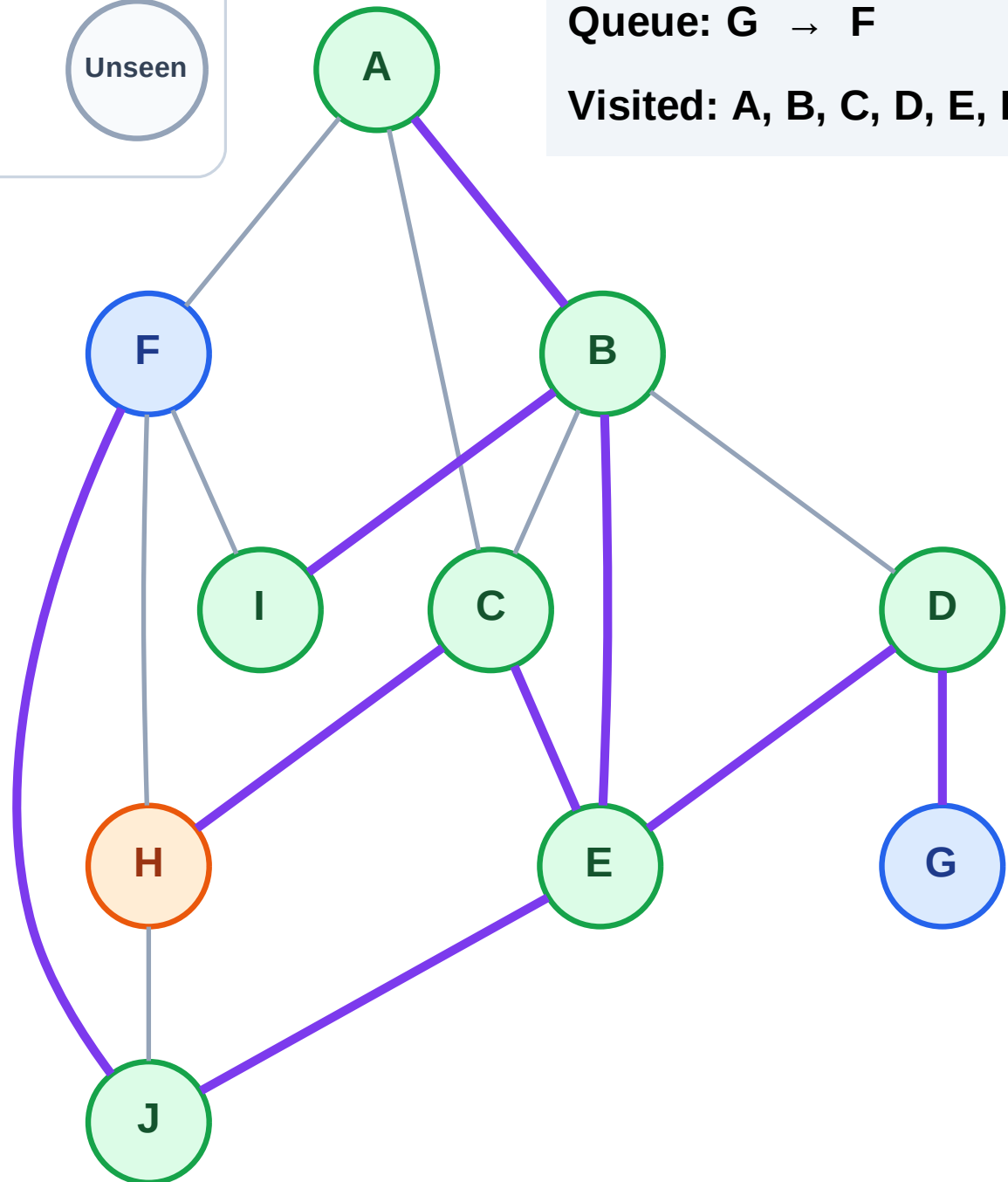
Processing

Discovered

Unseen

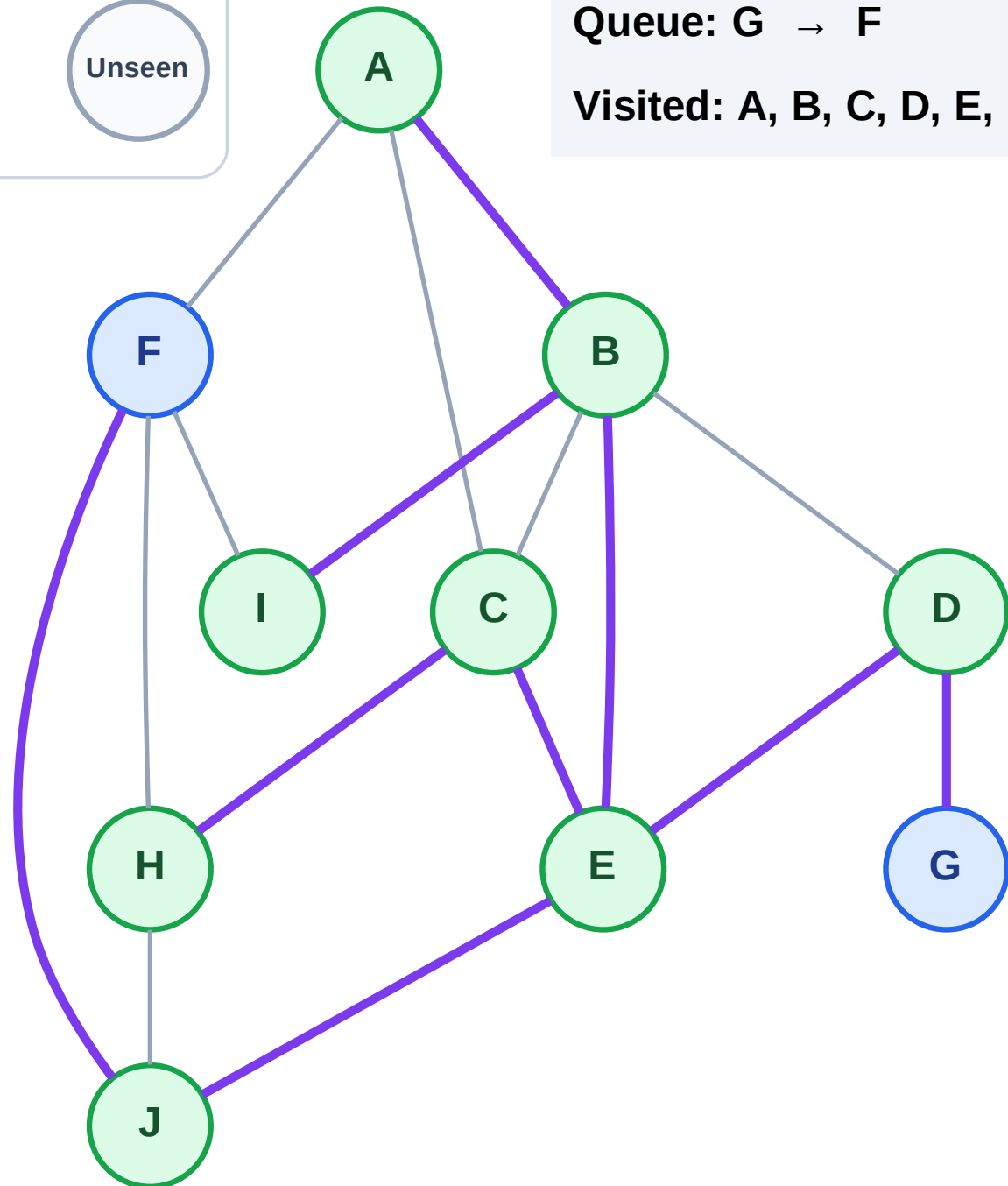
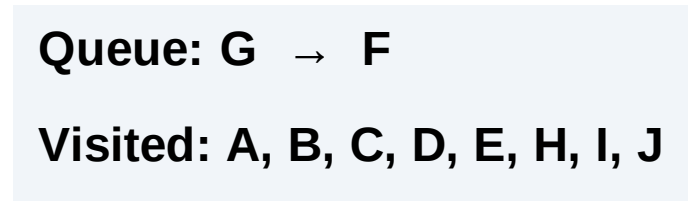
Queue: G → F

Visited: A, B, C, D, E, I, J



Step 17: No new neighbors from H

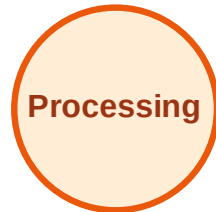
Step 17: No new neighbors from H



BFS Traversal

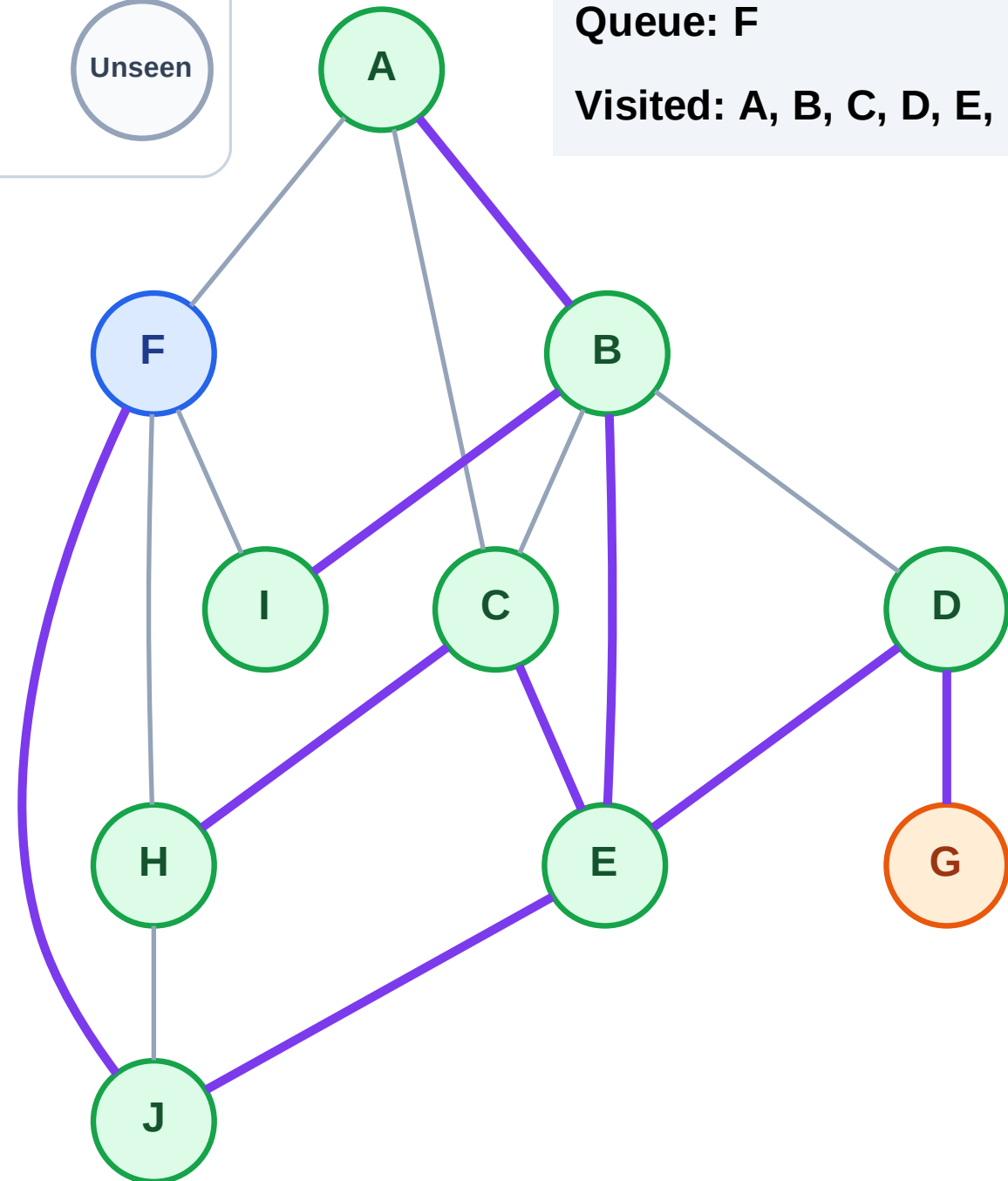
Step 18: Process node G

Legend



Queue: F

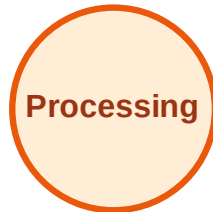
Visited: A, B, C, D, E, H, I, J



BFS Traversal

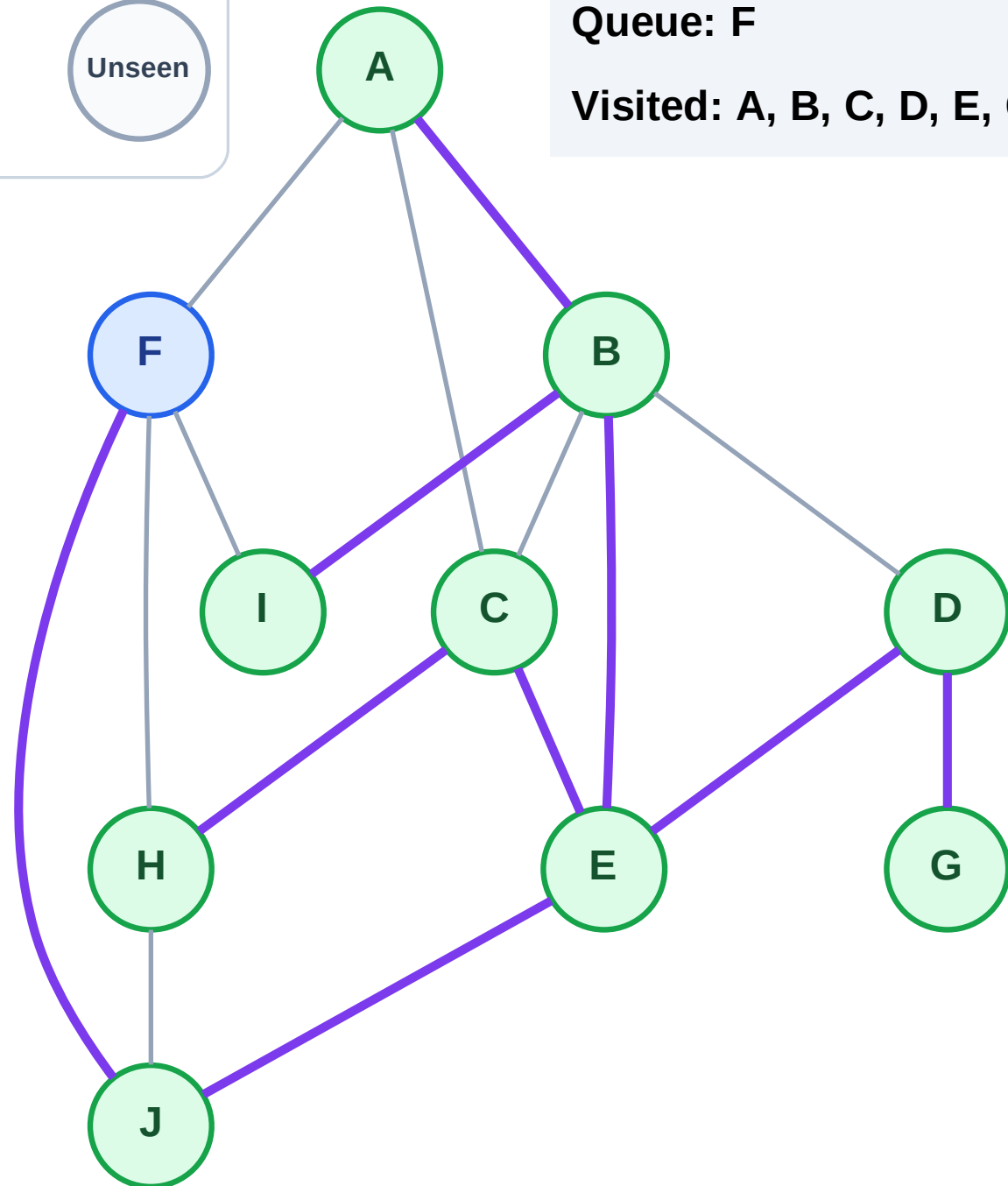
Step 19: No new neighbors from G

Legend



Queue: F

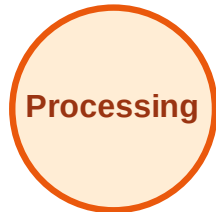
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

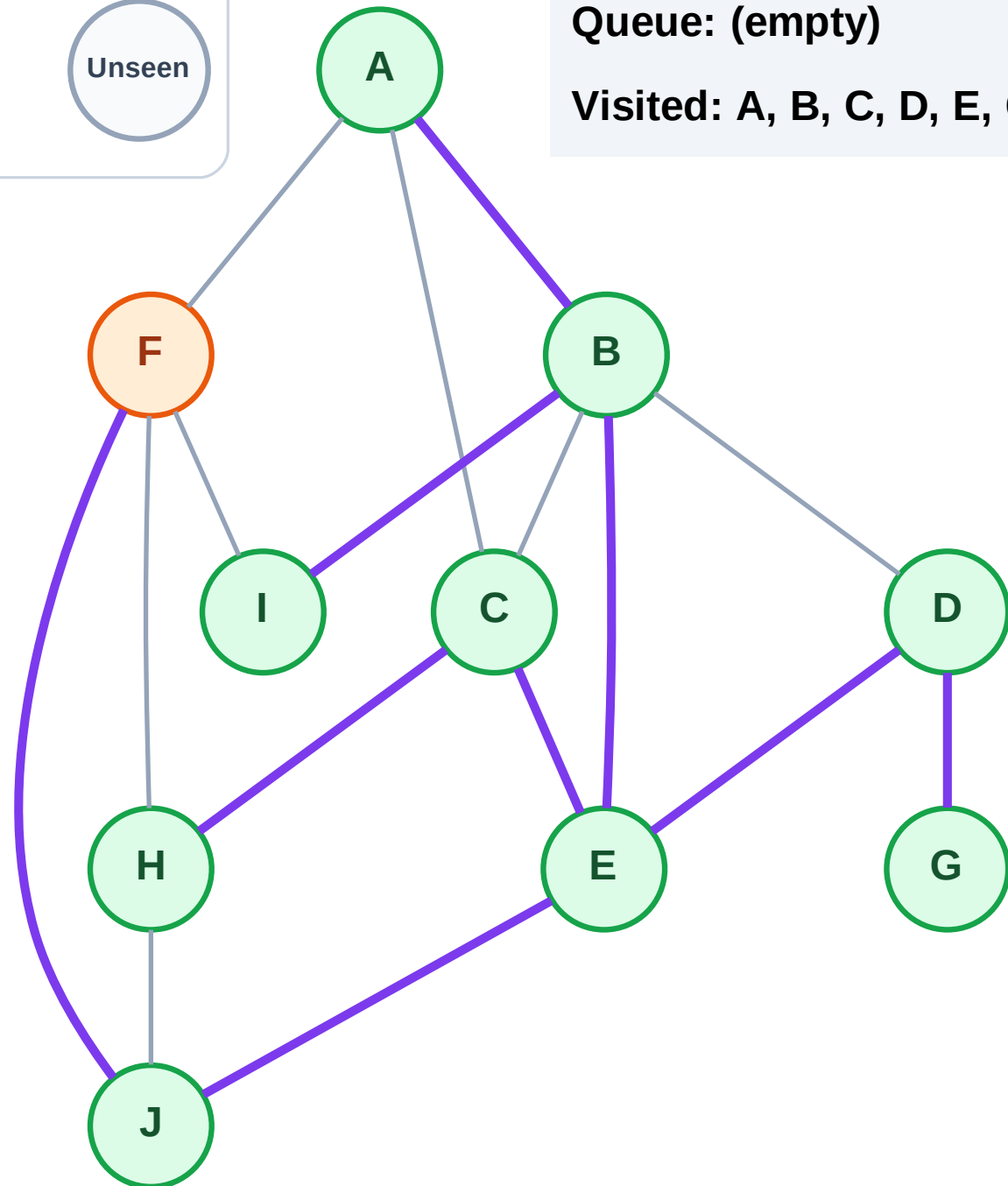
Step 20: Process node F

Legend



Queue: (empty)

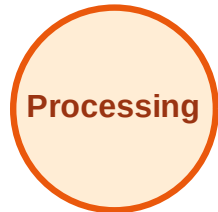
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

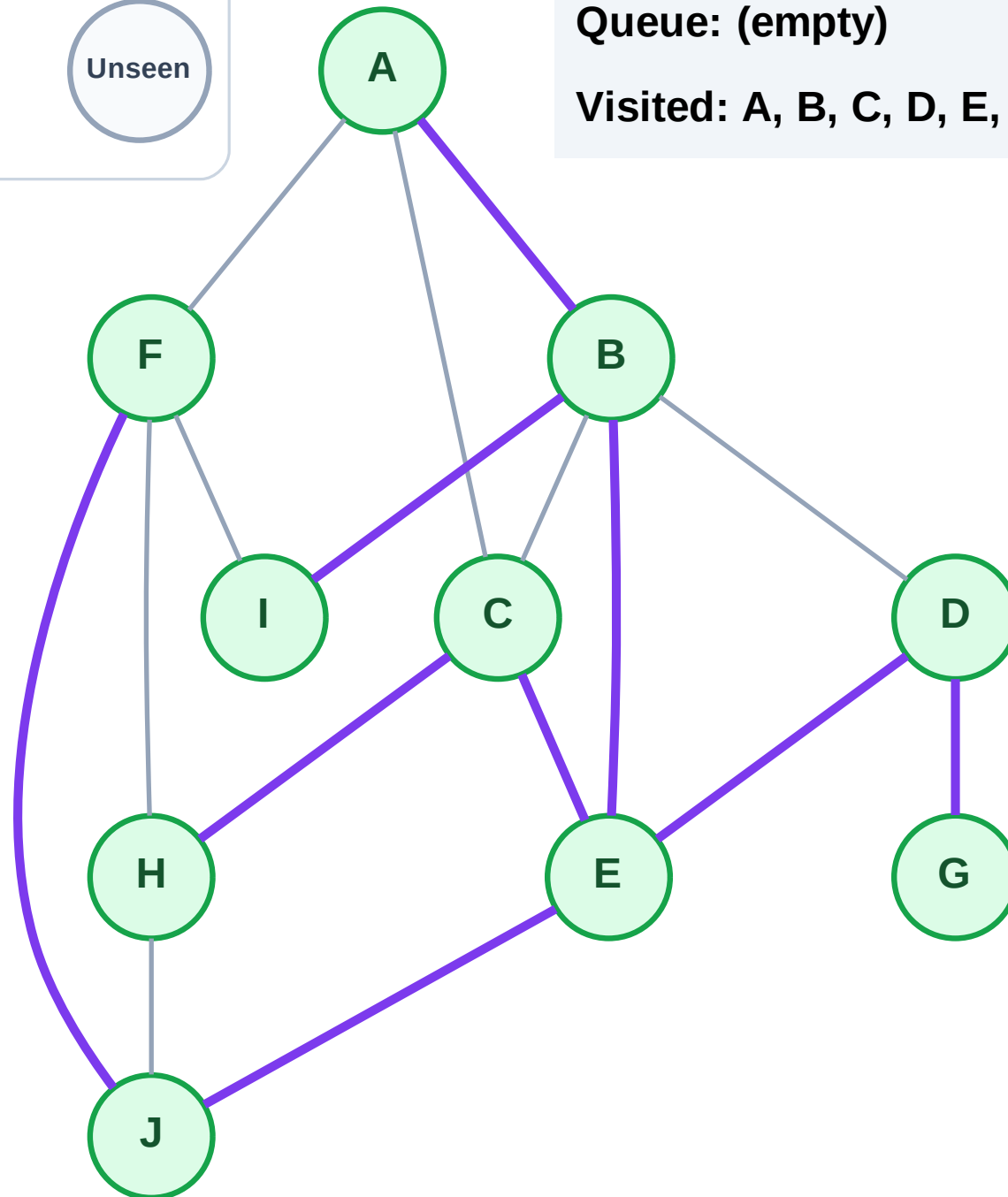
Step 21: No new neighbors from F

Legend



Queue: (empty)

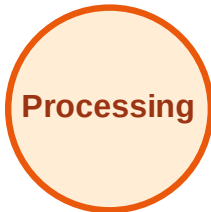
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

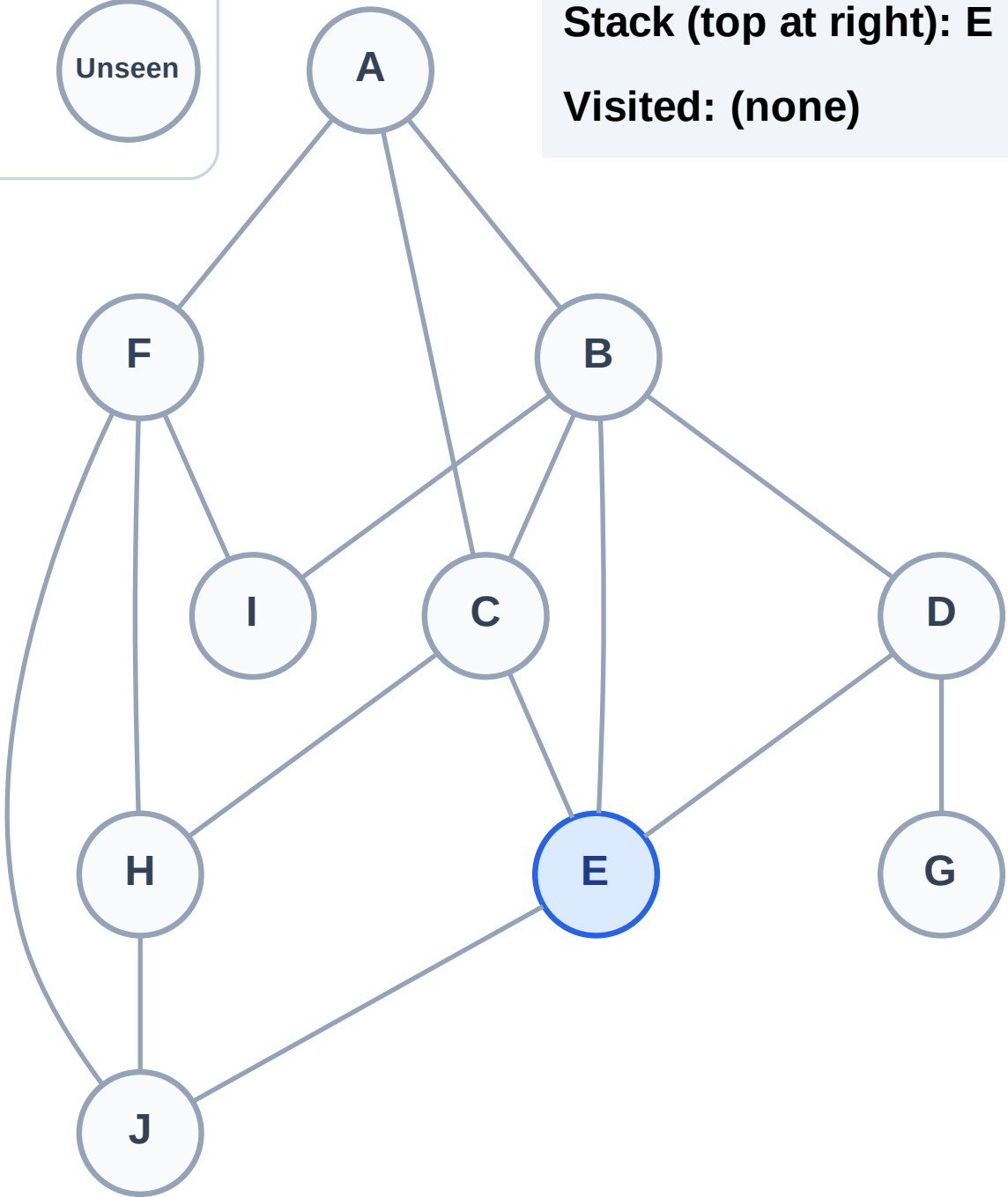
Step 1: Start at E

Legend



Stack (top at right): E

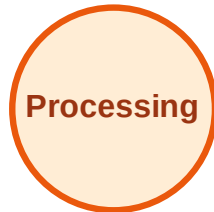
Visited: (none)



DFS Traversal

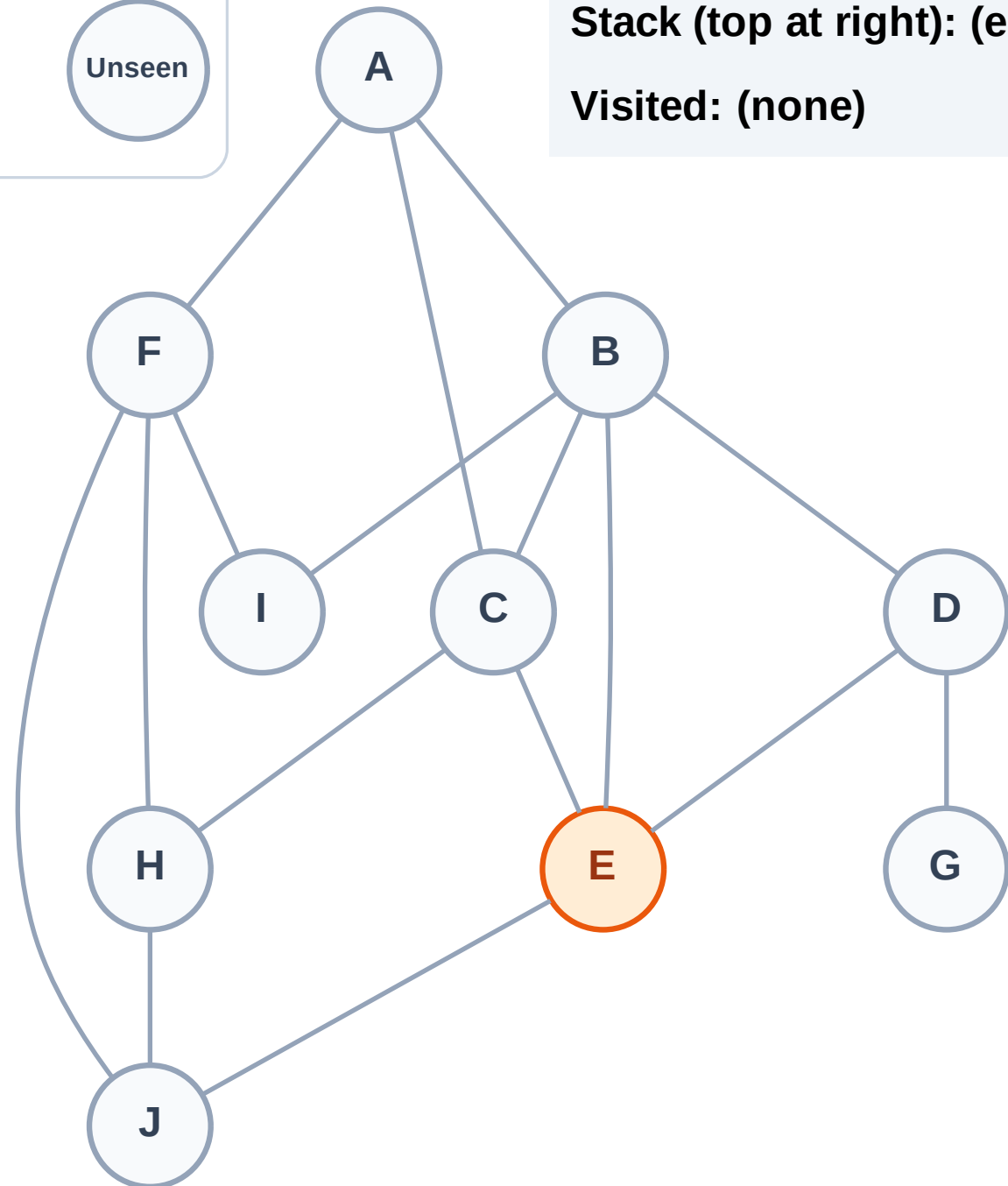
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover J, D, C, B from E

Legend

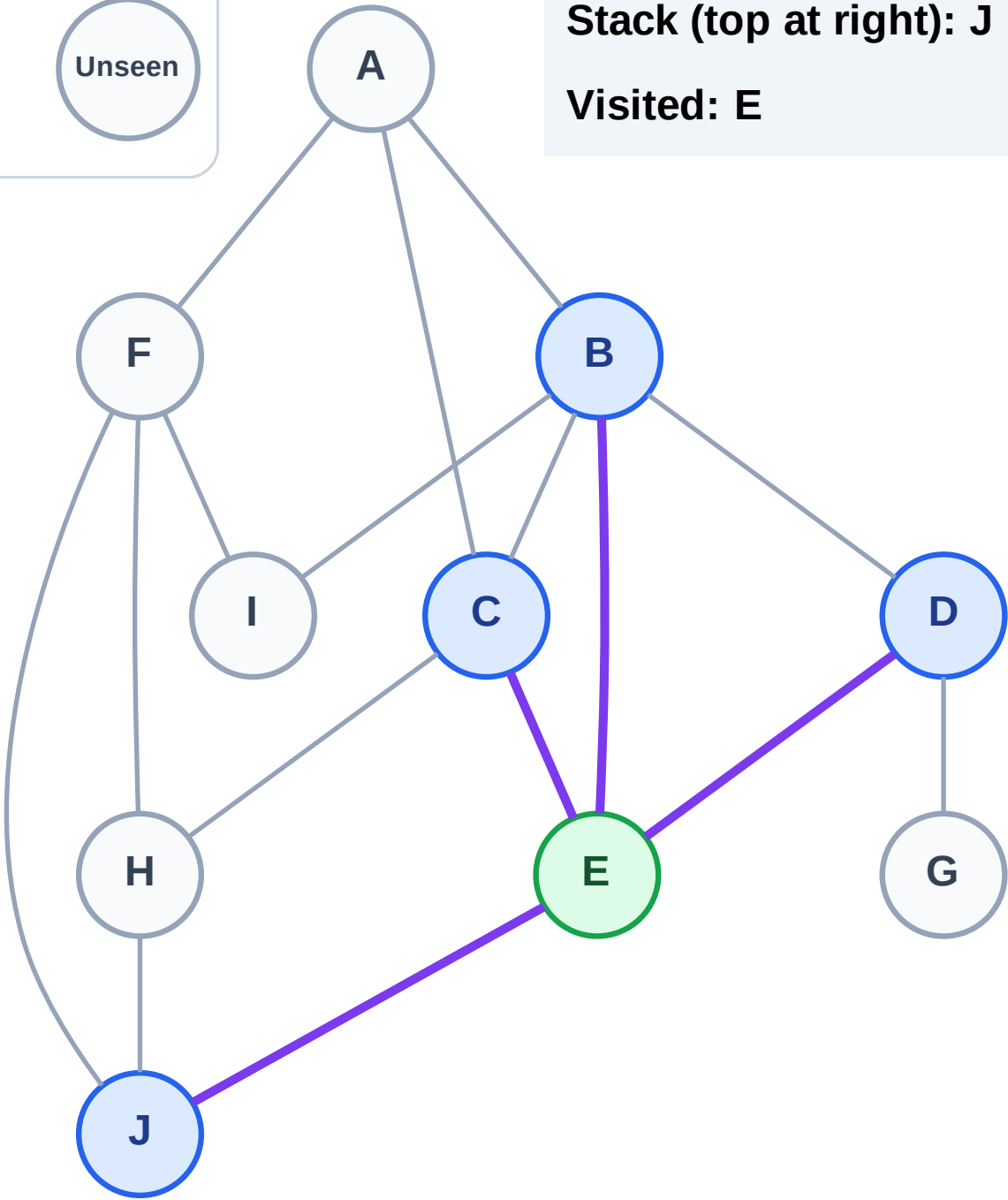
Complete

Processing

Discovered

Unseen

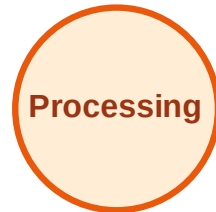
Stack (top at right): J | D | C | B
Visited: E



DFS Traversal

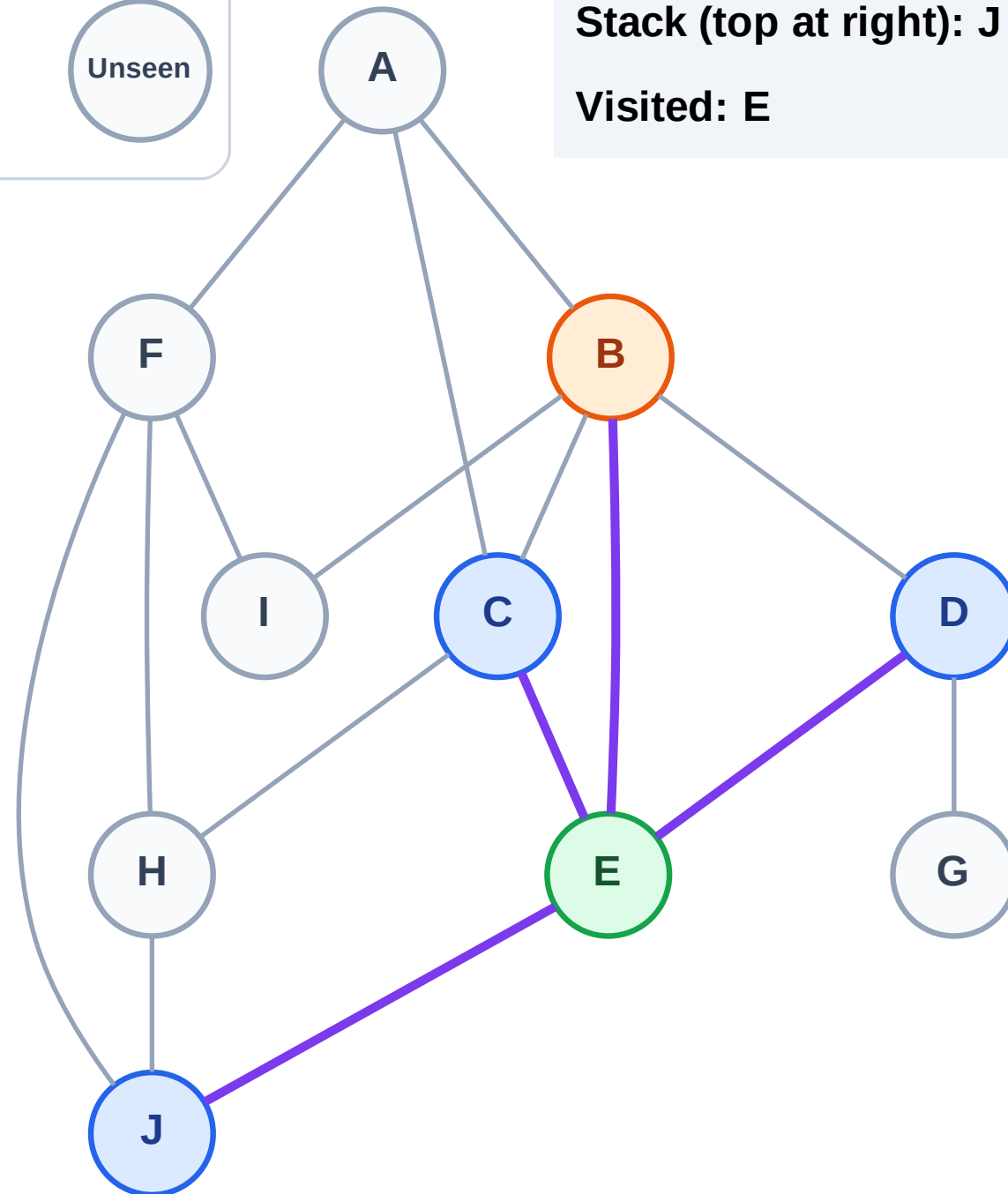
Step 4: Process node B

Legend



Stack (top at right): J | D | C

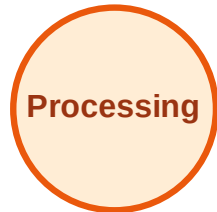
Visited: E



DFS Traversal

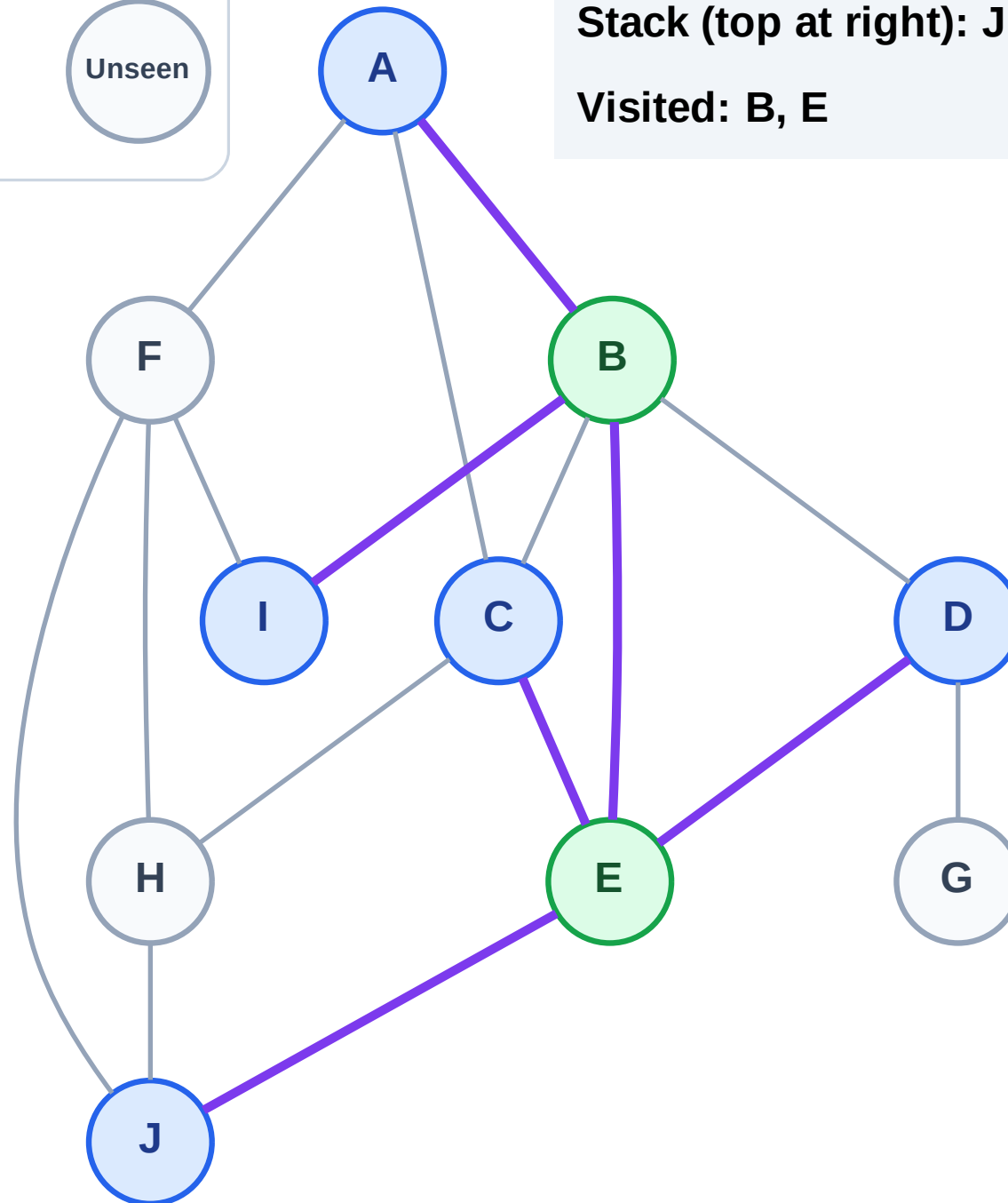
Step 5: Discover I, A from B

Legend



Stack (top at right): J | D | C | I | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

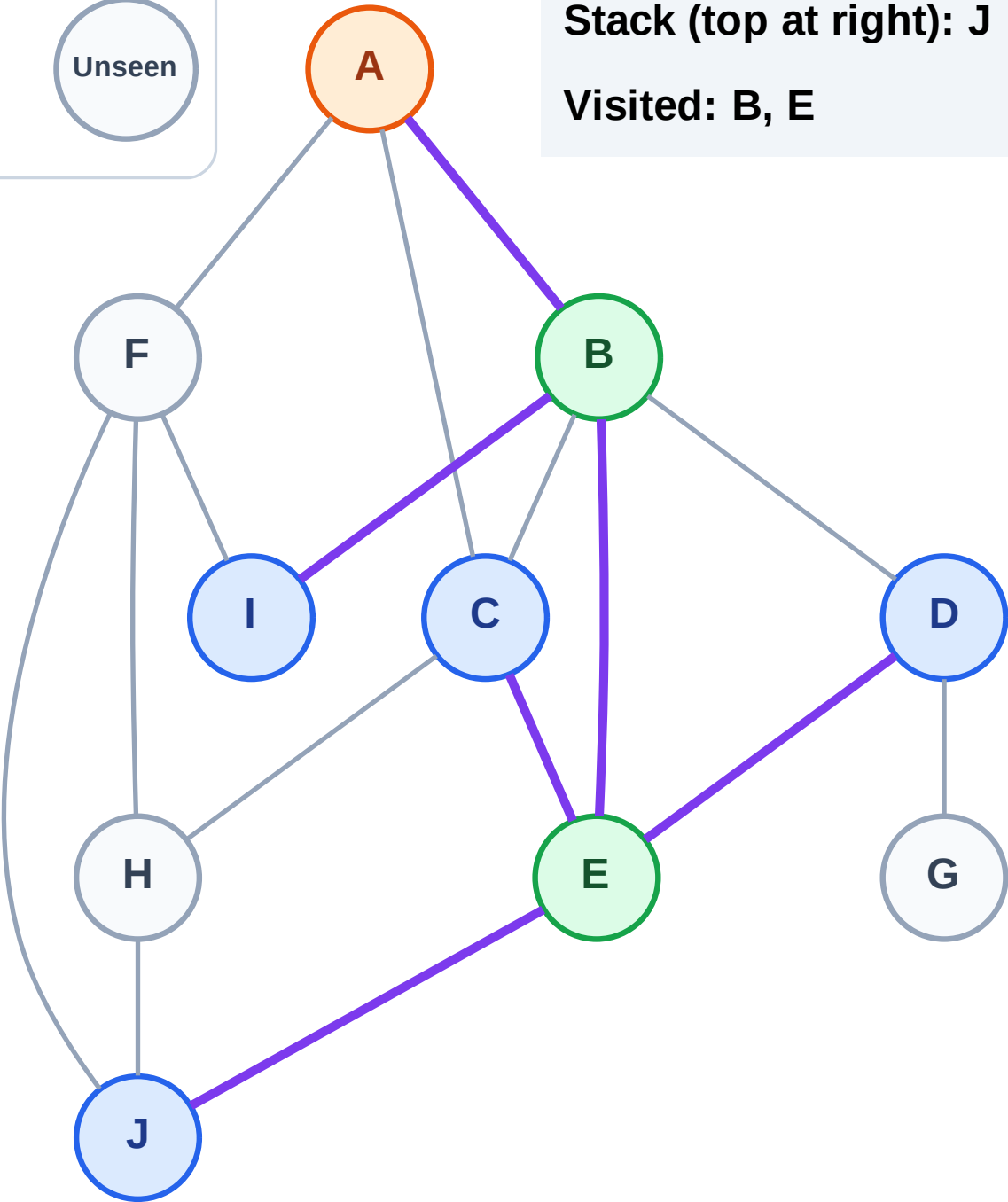
Complete

Processing

Discovered

Unseen

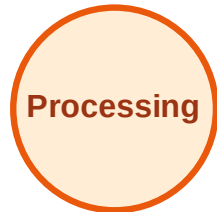
Stack (top at right): J | D | C | I
Visited: B, E



DFS Traversal

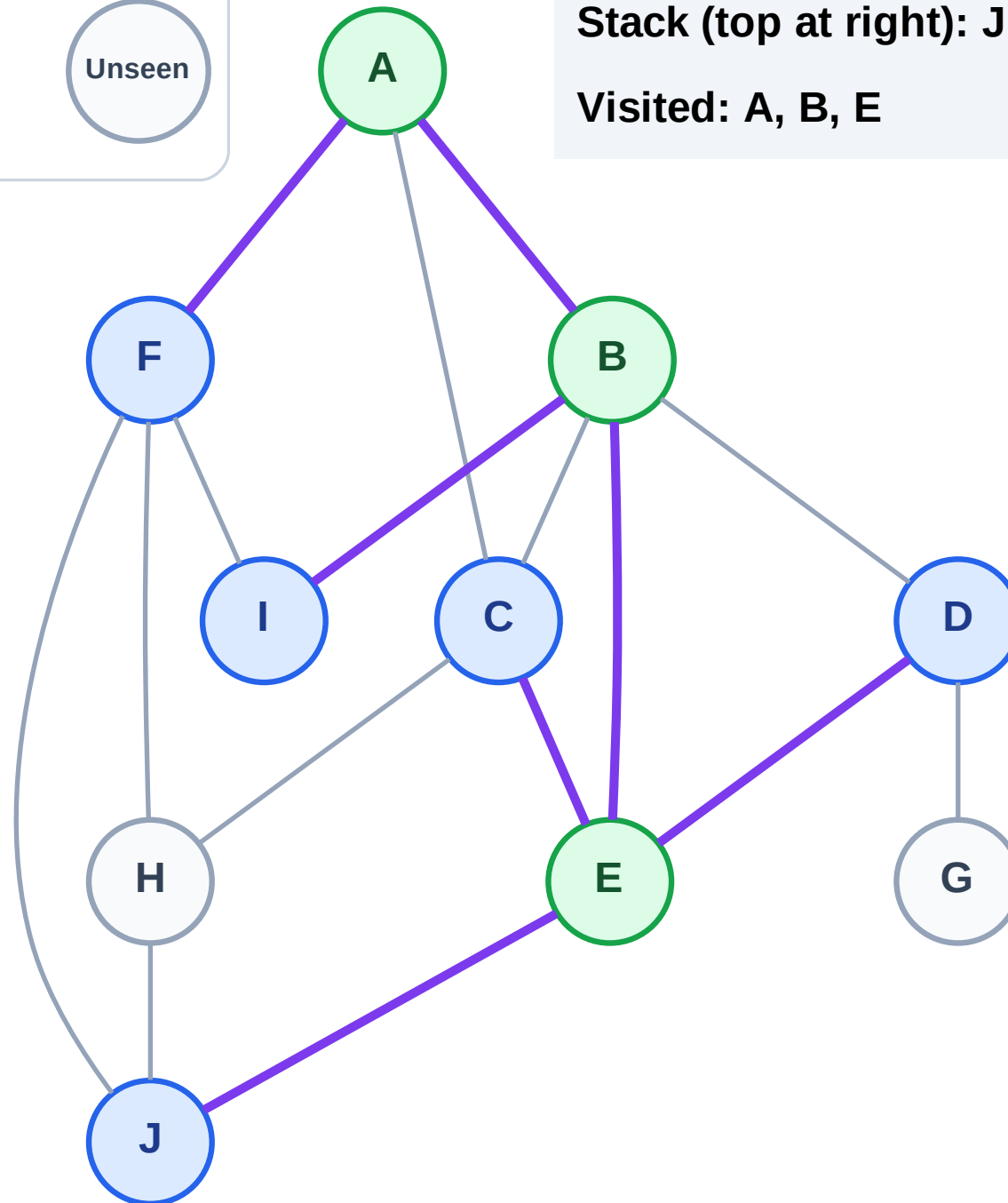
Step 7: Discover F from A

Legend



Stack (top at right): J | D | C | I | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

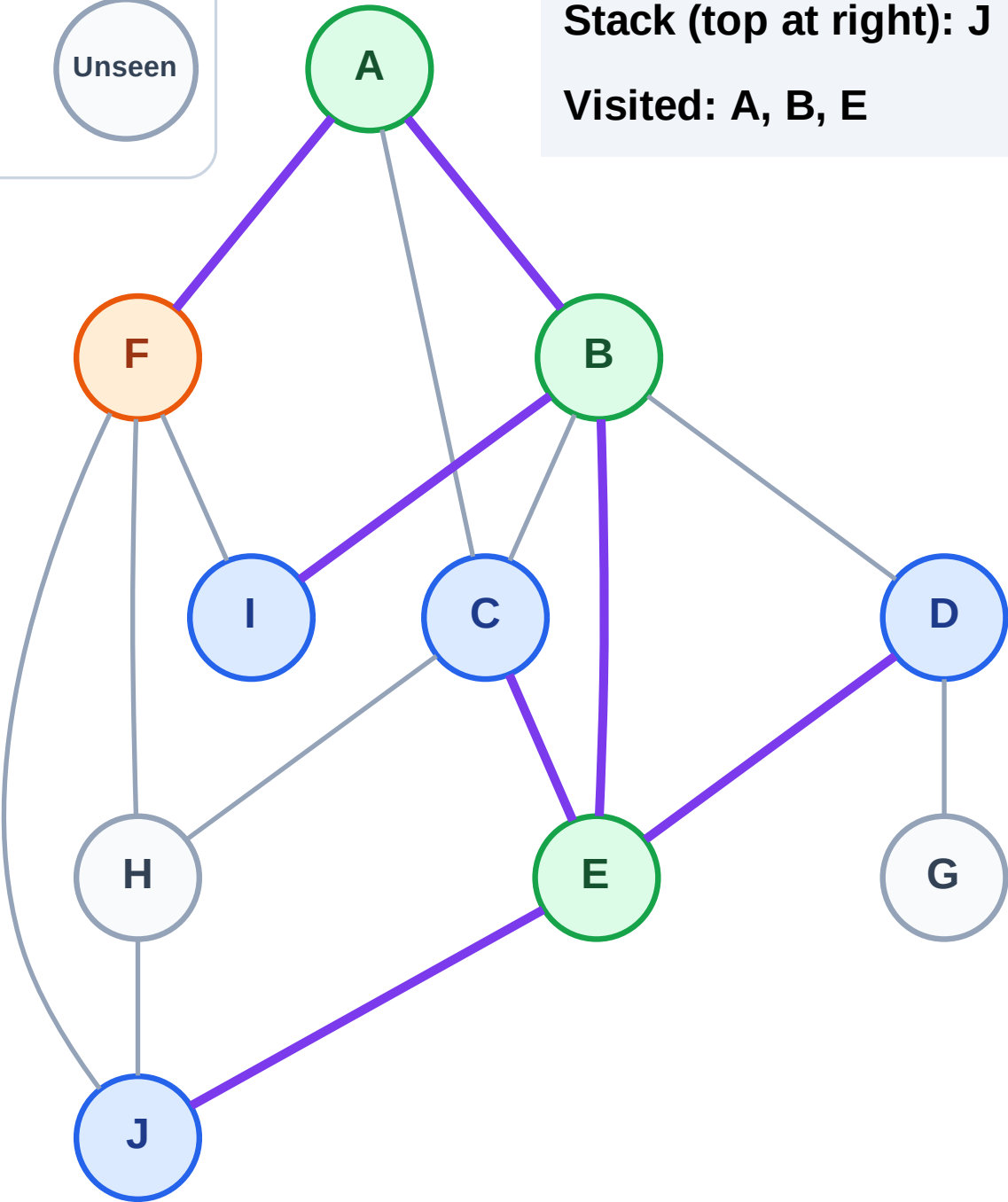
Processing

Discovered

Unseen

Stack (top at right): J | D | C | I

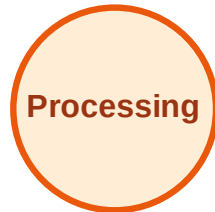
Visited: A, B, E



DFS Traversal

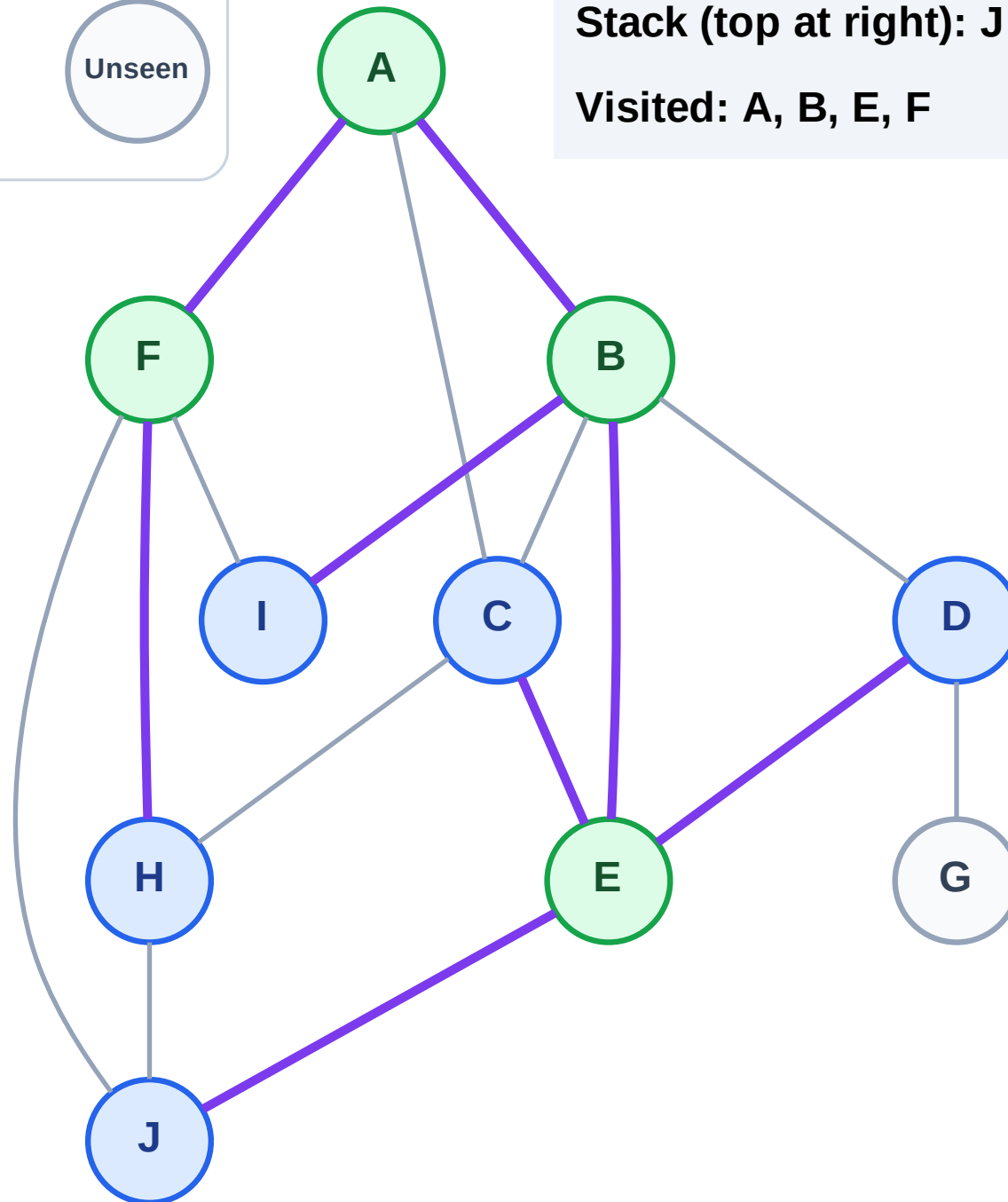
Step 9: Discover H from F

Legend



Stack (top at right): J | D | C | I | H

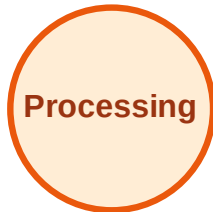
Visited: A, B, E, F



DFS Traversal

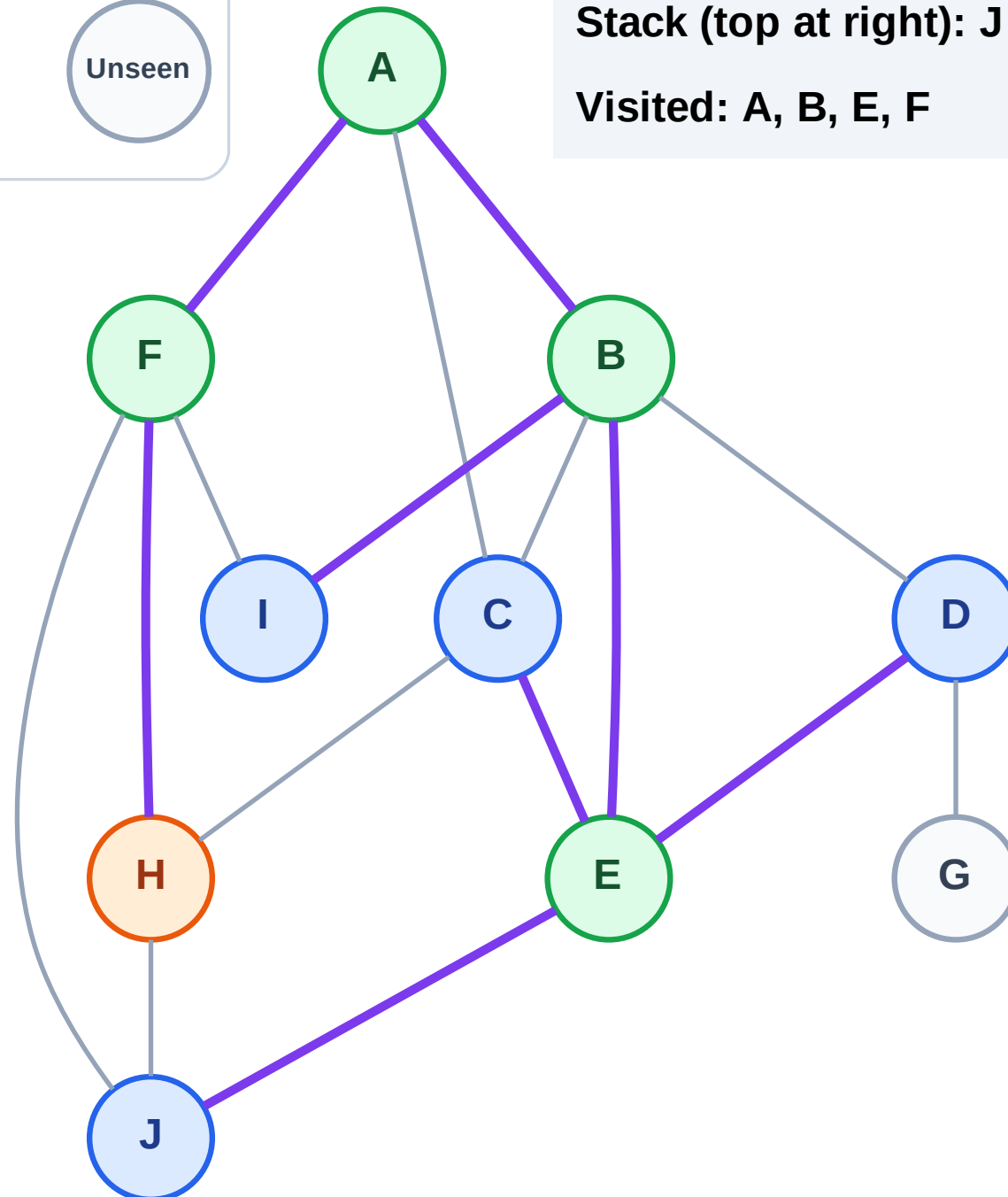
Step 10: Process node H

Legend



Stack (top at right): J | D | C | I

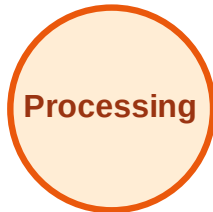
Visited: A, B, E, F



DFS Traversal

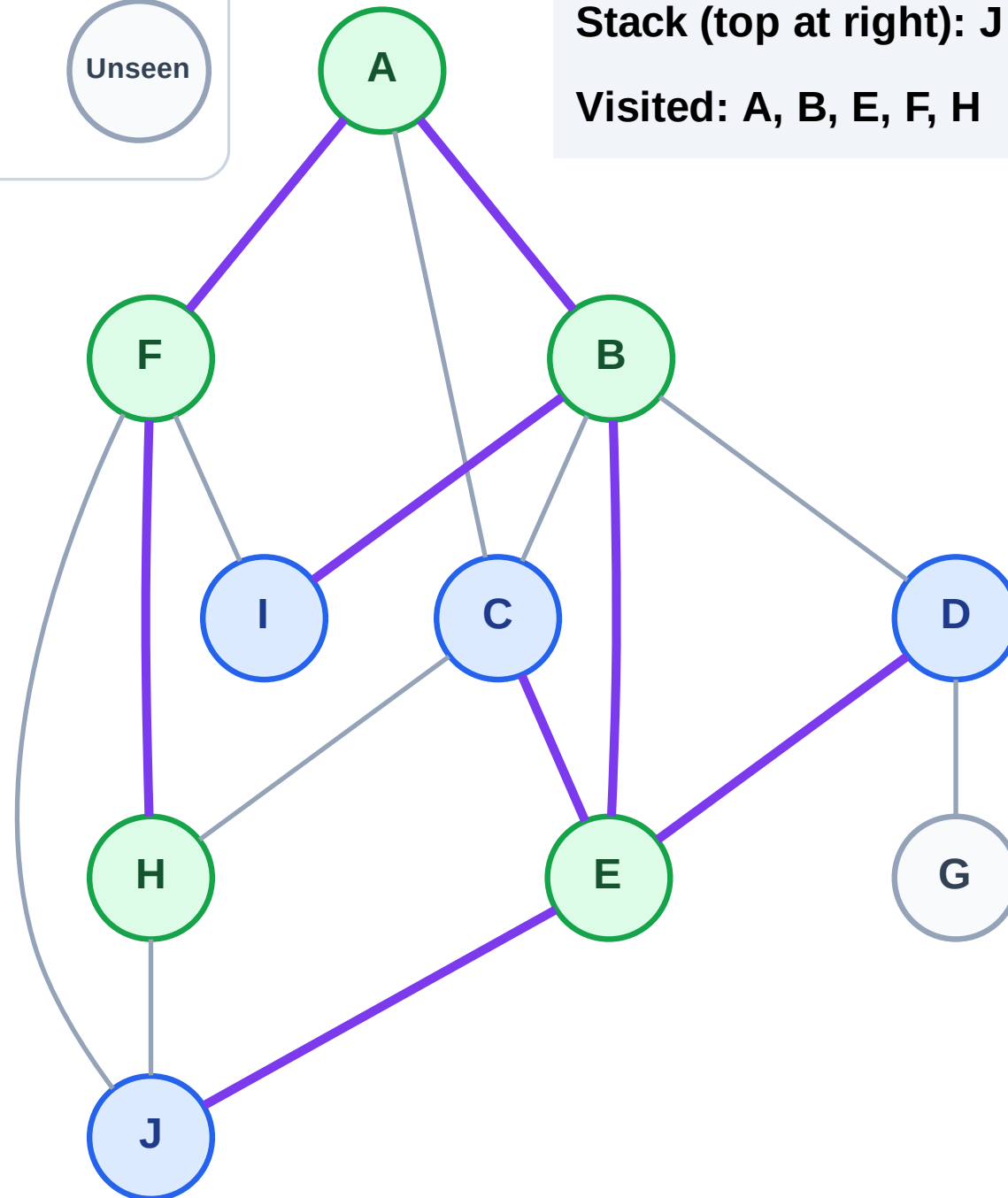
Step 11: No new neighbors from H

Legend



Stack (top at right): J | D | C | I

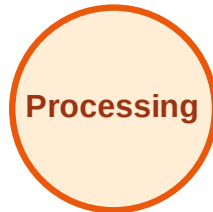
Visited: A, B, E, F, H



DFS Traversal

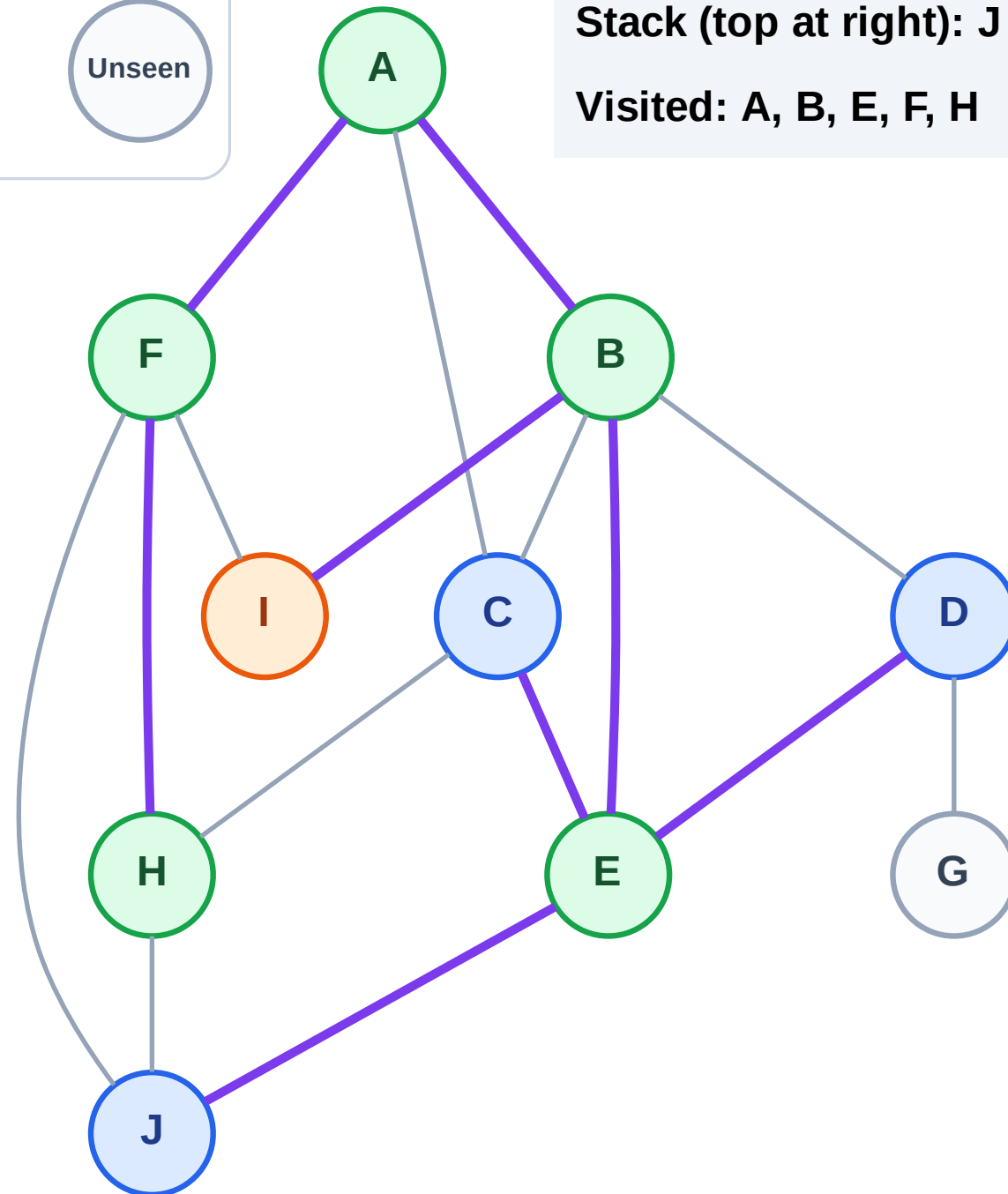
Step 12: Process node I

Legend



Stack (top at right): J | D | C

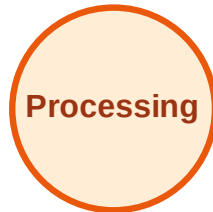
Visited: A, B, E, F, H



DFS Traversal

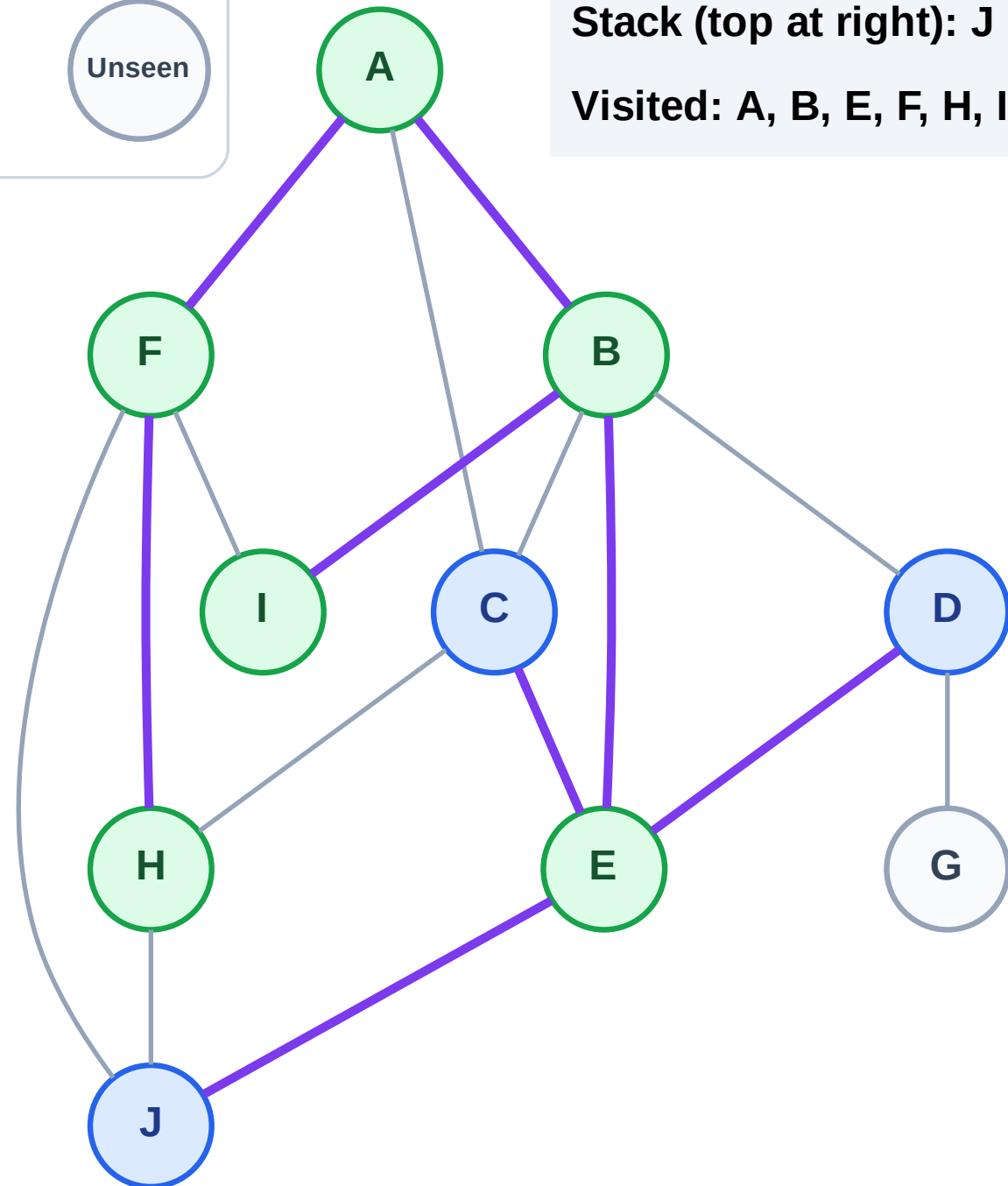
Step 13: No new neighbors from I

Legend



Stack (top at right): J | D | C

Visited: A, B, E, F, H, I



DFS Traversal

Step 14: Process node C

Legend

Complete

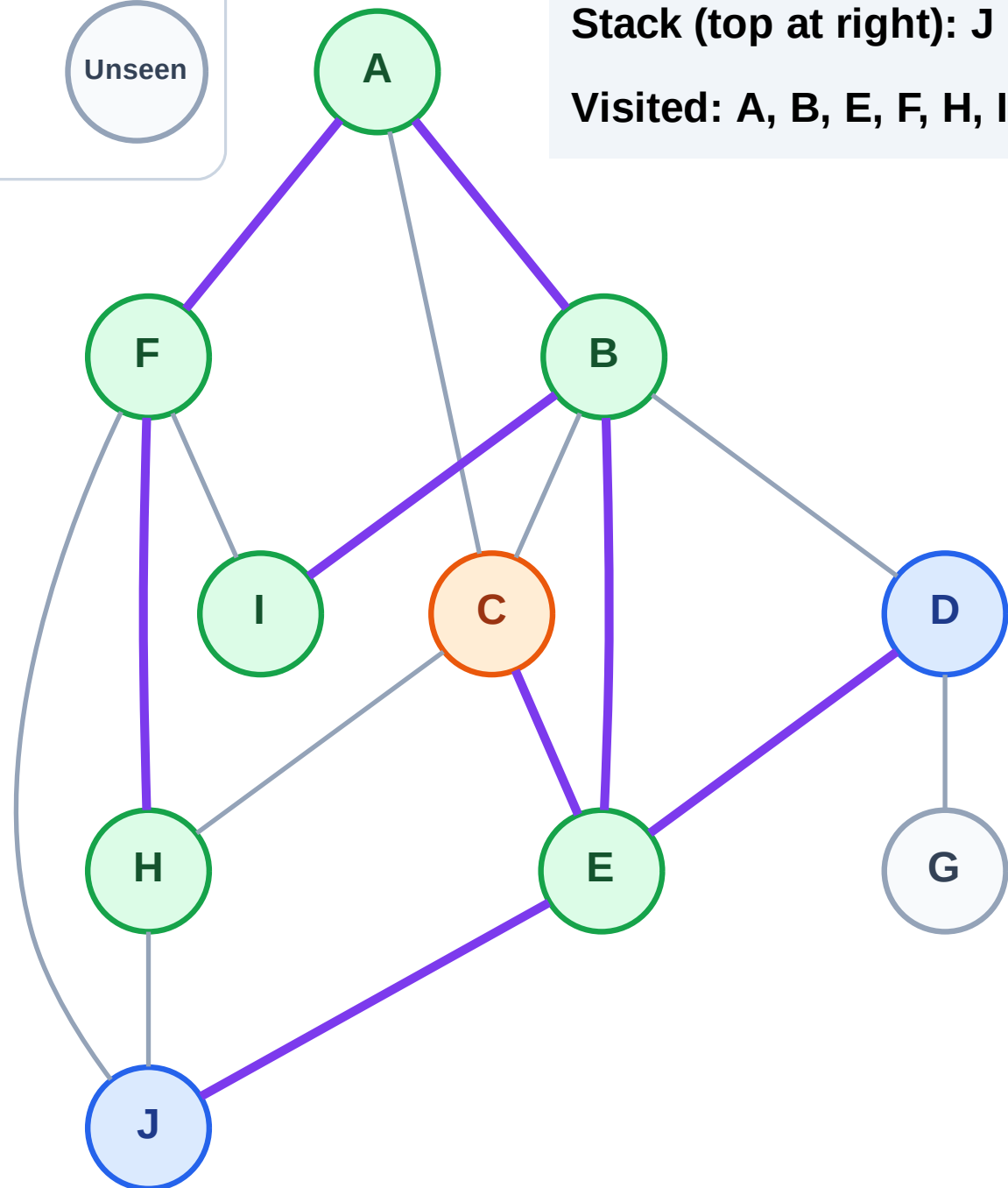
Processing

Discovered

Unseen

Stack (top at right): J | D

Visited: A, B, E, F, H, I



DFS Traversal

Step 15: No new neighbors from C

Legend

Complete

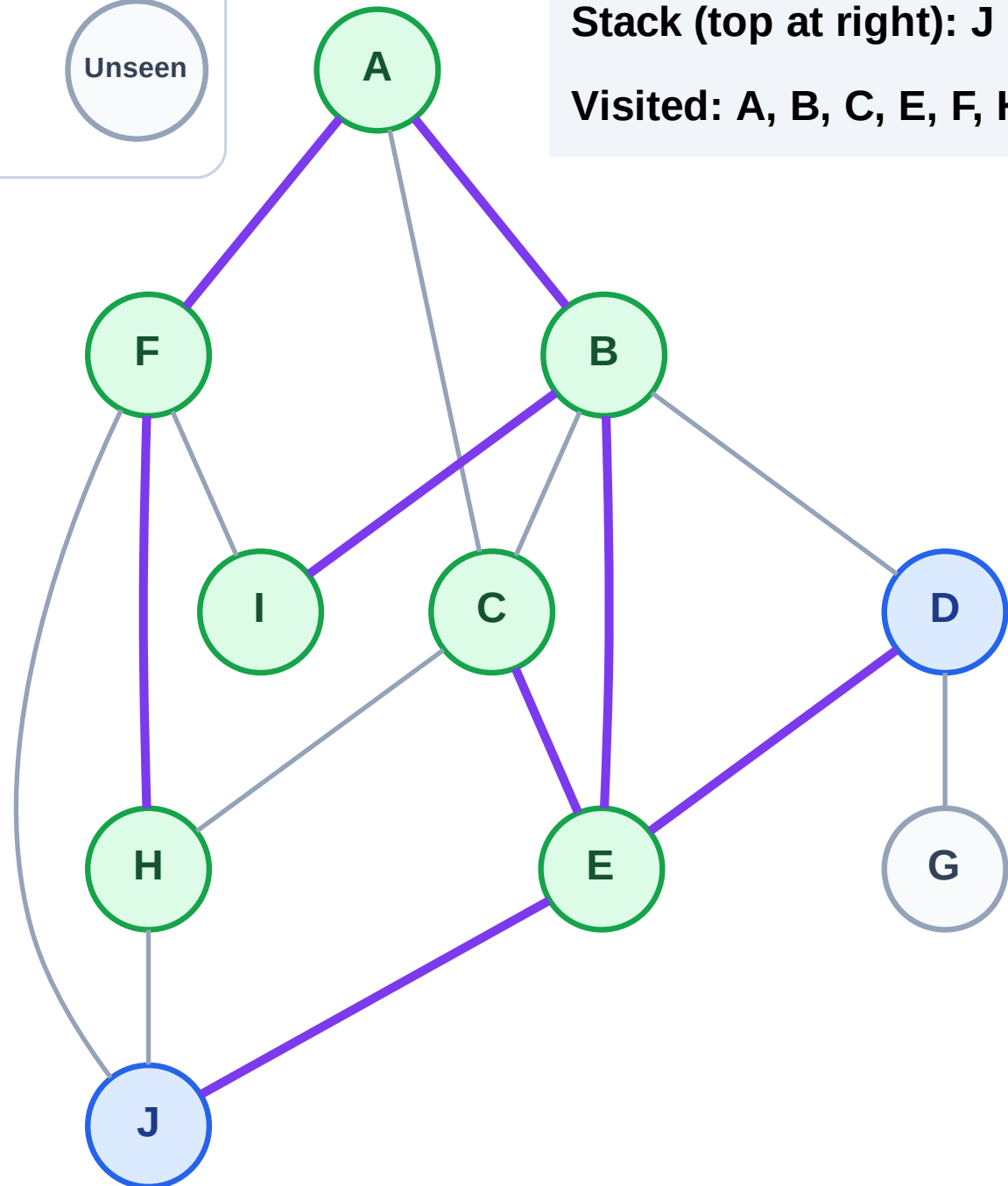
Processing

Discovered

Unseen

Stack (top at right): J | D

Visited: A, B, C, E, F, H, I



DFS Traversal

Step 16: Process node D

Legend

Complete

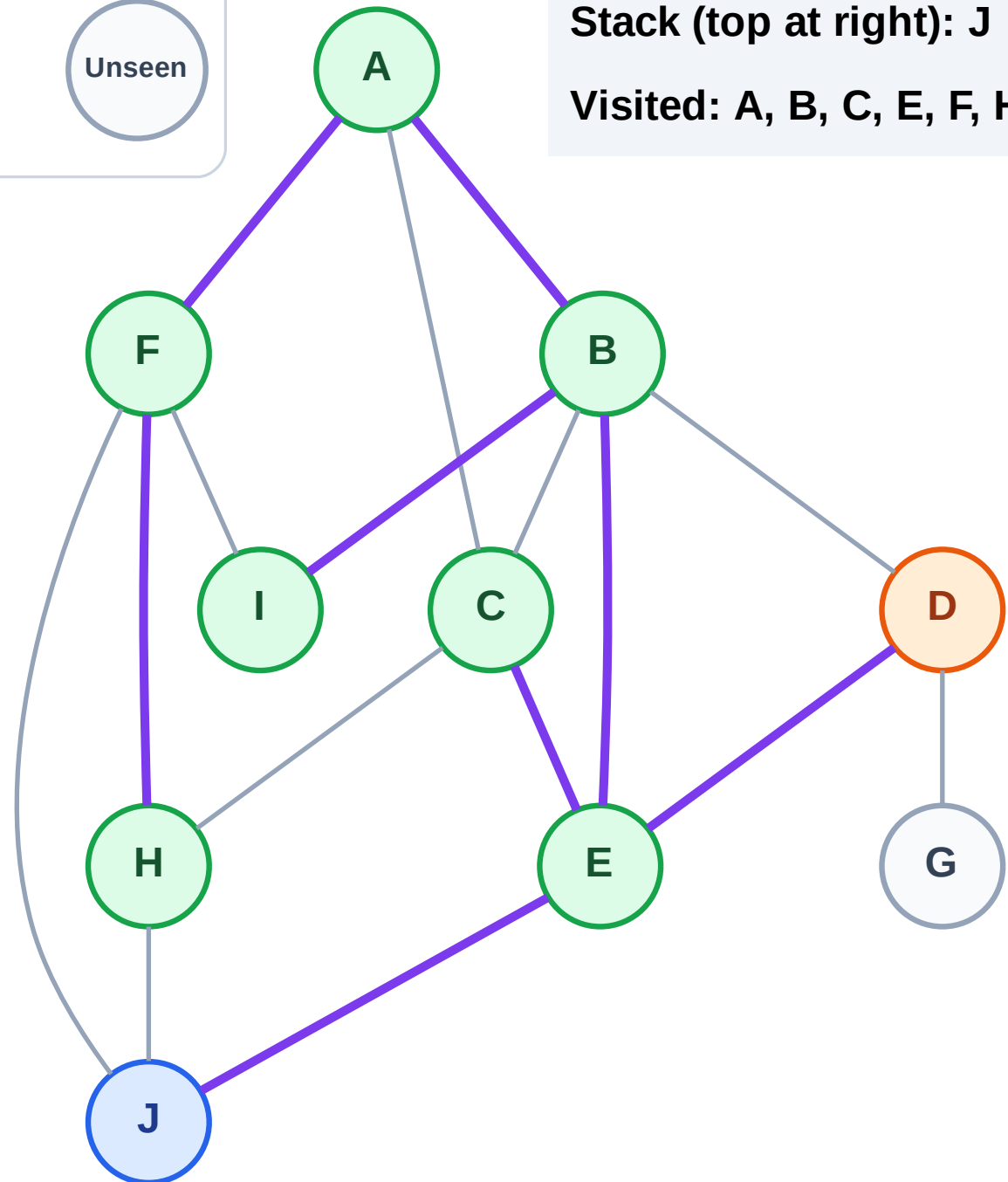
Processing

Discovered

Unseen

Stack (top at right): J

Visited: A, B, C, E, F, H, I



DFS Traversal

Step 17: Discover G from D

Legend

Complete

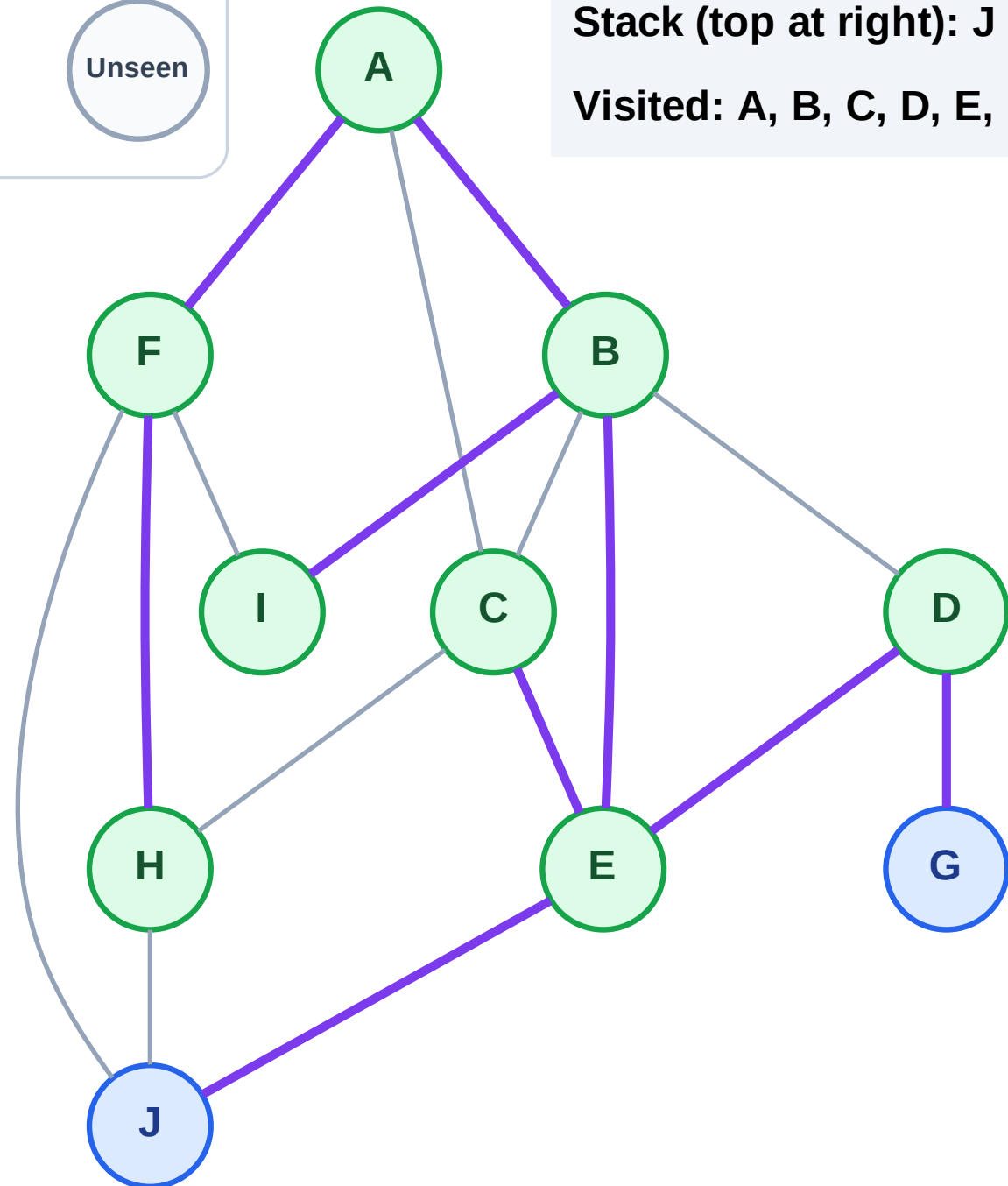
Processing

Discovered

Unseen

Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node G

Legend

Complete

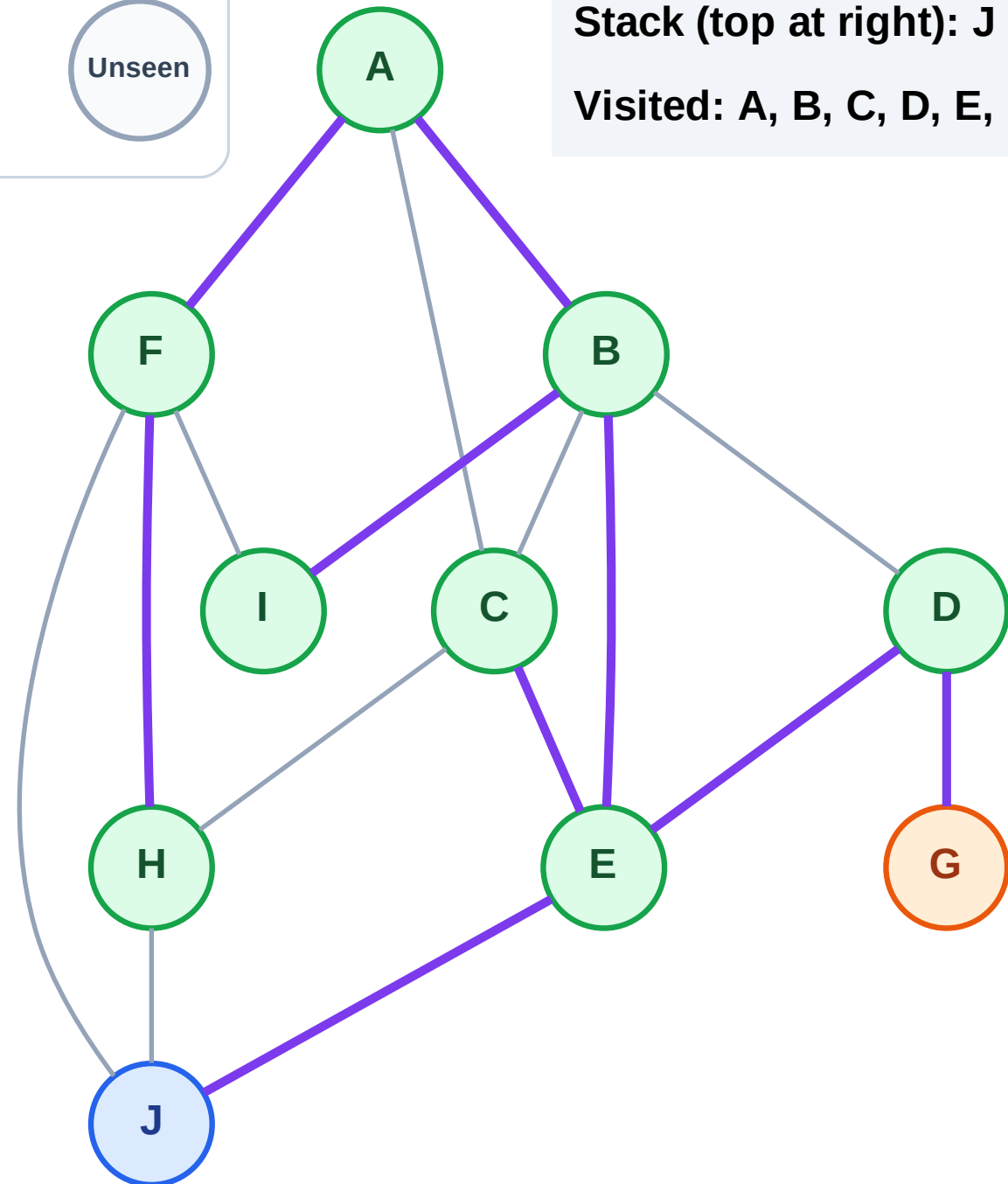
Processing

Discovered

Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, H, I

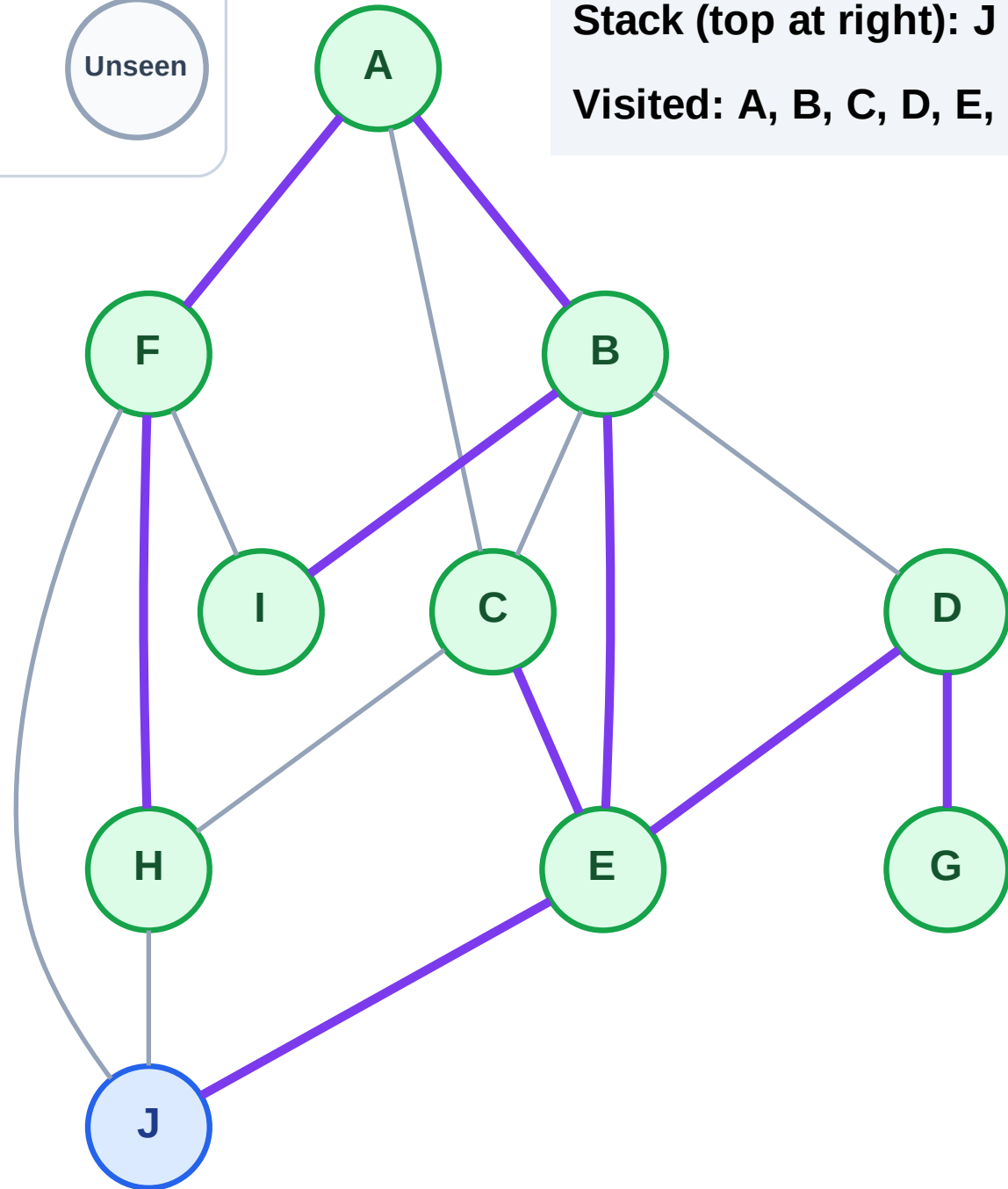


Step 19: No new neighbors from G

Legend



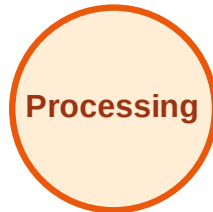
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

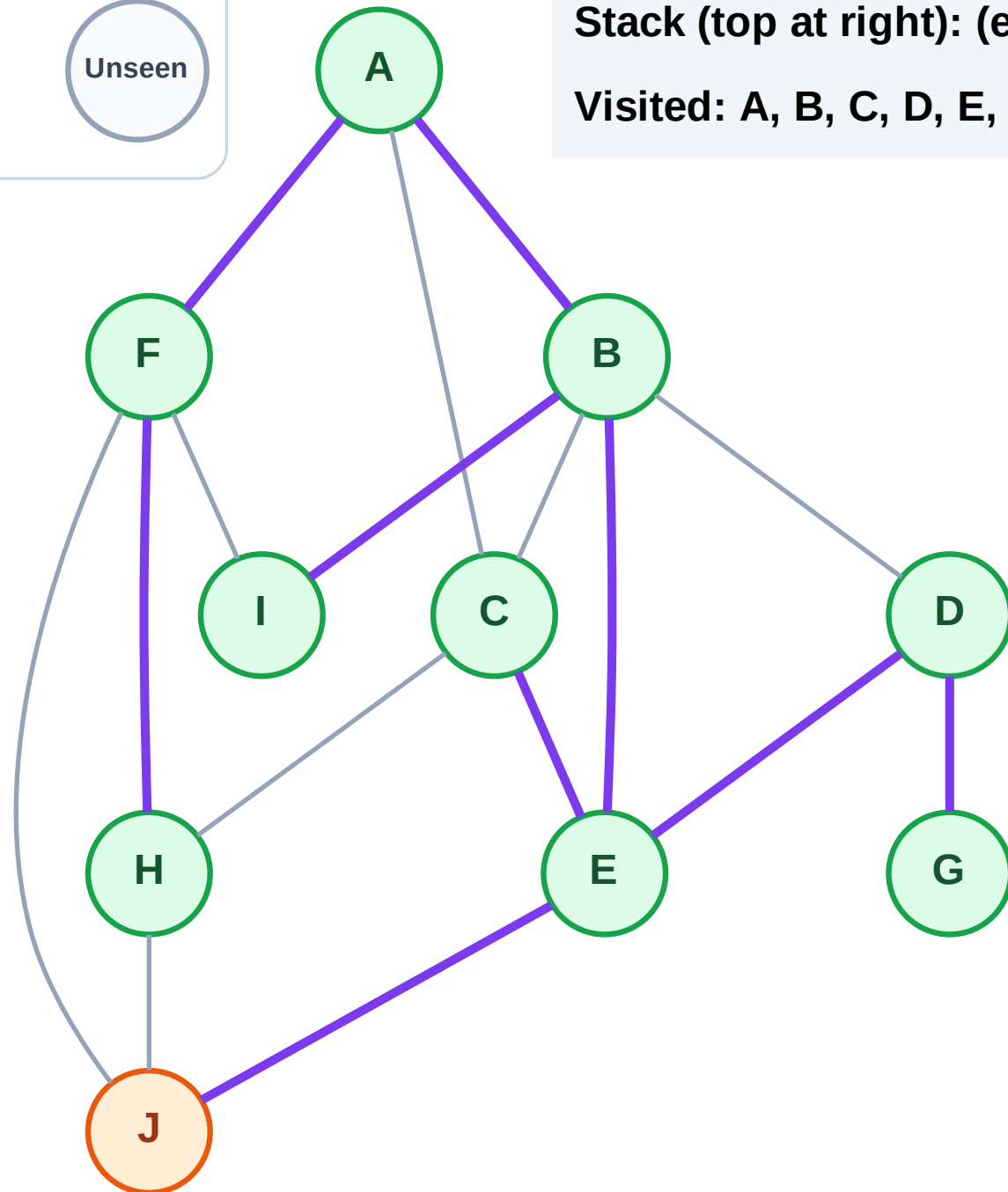
Step 20: Process node J

Legend



Stack (top at right): (empty)

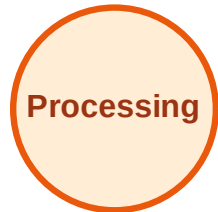
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

